



NI-MWMOTE: An improving noise-immunity majority weighted minority oversampling technique for imbalanced classification problems

Jianan Wei^a, Haisong Huang^{a,*}, Liguao Yao^{a,b}, Yao Hu^{a,c}, Qingsong Fan^a, Dong Huang^a

^a Key Laboratory of Advanced Manufacturing Technology, Ministry of Education, Guizhou University, Guiyang, Guizhou 550025, China

^b Department of Industrial, Engineering and Management, Yuan Ze University, Taoyuan 32003, Taiwan

^c Guizhou Renhe Zhiyuan, Data Service Co., Ltd., Guiyang, Guizhou 50025, China

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ABSTRACT

Oversampling techniques have been favored by researchers because of their simplicity and versatility in dealing with imbalanced classification problems. For oversampling techniques appeared in recent years (e.g. Majority Weighted Minority Oversampling Technique (MWMOTE)), noise processing plays an important role. This is because the processing of noise directly affects the distribution of new synthetic instances. MWMOTE and many other oversampling techniques use knn based noise processing method. While the knn method can effectively handle partial noise when the neighborhood parameter k value is reasonable, it may lead to under-recognition or over-recognition without prior experience. Therefore, we propose an improving noise-immunity majority weighted minority oversampling technique abbreviated NI-MWMOTE. NI-MWMOTE uses an adaptive noise processing scheme, which combines Euclidean distance and neighbor density to rank the probability that suspected noise (knn method) is true noise, and then adaptively selects the best noise processing scheme through iteration and misclassification error. Then, aggregative hierarchical clustering (AHC) method is used to cluster minority instances. And, in each sub-cluster, the sampling size of new samples is adaptively determined by classification complexity and cross-validation. NI-MWMOTE not only avoids the generation of new noise, but also effectively overcomes both between-class imbalances and within-class imbalances. Results demonstrate that NI-MWMOTE achieves significantly better results in most imbalanced datasets than eight popular oversampling algorithms.

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1. Introduction

With the rapid development of information technologies, historical data in various application fields has grown exponentially. However, these collected datasets are mostly imbalanced. That is, the number of instances in some categories is much less than the number of instances in others (Douzas, Bacao, & Last, 2018; Guo et al., 2017; Nekooimehr & Lai-Yuen, 2016), and an imbalance ratio is as high as 1/10000 or even higher. In fact, the minority instances have more research value. This is because these data often contain more important information, such as customer churn prediction (Xiao, Xie, He, & Jiang, 2012), DNA microarray data analysis (Yu, Ni, & Zhao, 2013), software defect detection (Siers & Islam, 2018), mechanical fault diagnosis (Mao, He, Yan, & Wang, 2017)

and medical diagnosis (Han, Huang, Li, & Jia, 2019). Therefore, imbalanced classification problems are attracting more and more researchers' interest.

Traditional standard classifiers aim to maximize the accuracy of overall classification. However, when dealing with imbalanced datasets, they tend to pay too much attention to majority instances and ignore minority instances, which will result in the loss of classifier performance (Prati, Batista, & Silva, 2015). When the dataset is severely imbalanced, classifiers may have an undoing effect on minority instances (Mao et al., 2017). This kind of imbalance that exists between two different classes is called between-class imbalance. However, in practical applications, most of the imbalanced datasets have another imbalance, called within-class imbalance (Barua, Islam, Yao, & Murase, 2014; Douzas et al., 2018; Nekooimehr & Lai-Yuen, 2016). Within-class imbalance occurs when the minority (majority) instances have more than one sub-clusters and some of these sub-clusters have less number of instances than others. Additionally, the overlap of instances between different classes brings another great challenge to the

* Corresponding author.

E-mail addresses: jianan_wei_gzu@163.com (J. Wei), hshuang@gzu.edu.cn (H. Huang), yaoliguao1990@163.com (L. Yao), yhu3@gzu.edu.cn (Y. Hu), qs_fan@126.com (Q. Fan), chenyijia000011@163.com (D. Huang).

imbalanced classification problems (Alshomrani, Bawakid, Shim, Fernandez, & Herrera, 2015). This is because the overlap of samples will blur the classification boundaries.

To address the above problems, many popular oversampling algorithms have been proposed. While oversampling algorithms (e.g. ROS (Random oversampling), SMOTE (Synthetic Minority Oversampling Technique) (Chawla, Bowyer, Hall, & Kegelmeyer, 2002), BSMOTE (Borderline-SMOTE) (H. Han, Wang, & Mao, 2005), ADASYN (adaptive synthetic sampling approach) (He, Yang, Garcia, & Li, 2008) and Safe-level-SMOTE (safe-level-synthetic minority oversampling technique) (Bunkhumpornpat, Sinapiromsaran, & Lursinsap, 2009)) can alleviate between-class imbalances to a certain extent, they can not alleviate within-class imbalances. Therefore, recently, many scholars have combined clustering algorithms with popular oversampling algorithms (e.g. SMOTE) to overcome both between-class imbalances and within-class imbalances. Typical algorithms in this regard include Cluster-SMOTE (Cieslak, Chawla, & Striegel, 2006), MWMOTE (majority weighted minority oversampling technique) (Barua et al., 2014), DBSMOTE (density-based synthetic minority oversampling technique) (C. Bunkhumpornpat, Sinapiromsaran, & Lursinsap, 2012), CURE-SMOTE (Ma & Fan, 2017), A-SUWO (adaptive semi-supervised weighted oversampling) (Nekooimehr & Lai-Yuen, 2016) and K-means SMOTE (Douzas et al., 2018), etc. While the above algorithms strive to address both between-class imbalances and within-class imbalances, they have the following shortcomings:

- 1) Some algorithms (e.g. Cluster SMOTE and DBSMOTE) introduce lots of new noise after trying to address within-class imbalances, which exacerbates the overlap between categories (Nekooimehr & Lai-Yuen, 2016).
- 2) Some algorithms (e.g. A-SUWO) have higher complexity and poor practicability (Douzas et al., 2018), and are prone to over-fitting.
- 3) Some algorithms (e.g. K-means SMOTE and CURE-SMOTE) only sample in the safe area, so it is easy to ignore important boundary instances. However, for imbalanced classification problems, those boundary instances that are difficult to learn are very important and cannot be ignored (Barua et al., 2014; Han et al., 2005; He et al., 2008; Nekooimehr & Lai-Yuen, 2016).
- 4) None of the above algorithms deal with noise reasonably. In fact, the processing of noise directly affects the distribution of new instances, that is, directly affects the sampling effect of the algorithm.
- 5) When a dataset is seriously imbalanced, minority instances are sparse and there is some overlap between different classes, some algorithms (e.g. A-SUWO and K-means SMOTE, etc.) are almost ineffective. (Regarding the explanation of 1)–5), we will show some scenarios in Section 2 to illustrate.)

Therefore, to address the above problems, we propose an improving noise-immunity majority weighted minority oversampling technique (NI-MWMOTE) to address the binary imbalanced classification problems. Unlike MWMOTE, NI-MWMOTE first adaptively removes noise. The process intends to increase the difference between true noise and pseudo noise by considering the Euclidean distance from an instance to the class center and its nearest neighbor density. And then, it calculate the probabilities that the suspected noise (knn method) is regarded as true noise. Next, based on these probabilities, it uses iteration and misclassification error of original dataset to determine the best noise processing scheme. Then, NI-MWMOTE uses AHC to segment minority instances. There are two advantages to this operation: 1) It is not easy to ignore

small minority sub-clusters that are far away from the majority instances. This will increase the probabilities that all boundary instances can be sampled. 2) It will not greatly increase the complexity of NI-MWMOTE. Finally, NI-MWMOTE can adaptively determine the number of instances that sub-clusters need to synthesize. Specifically, it first determines misclassification error by standardizing average error rate and cross-validation. And then, misclassification error is used to be a measure of sub-cluster complexity, which finally makes the smaller minority sub-clusters get more chances to be sampled. Shortly, NI-MWMOTE intends to overcome both between-class imbalances and within-class imbalances while reasonably processing noise. And it does not significantly increase the complexity of the algorithm.

To validate the effectiveness of NI-MWMOTE, an extensive experimental design is necessary. The proposed NI-MWMOTE algorithm is tested on 24 kinds of publicly available datasets, classified using four classifiers (SVM, ANN, RF and KNN) and compared with eight popular oversampling techniques, where Sensitivity, F-measure, G-mean and AUC are used as evaluation measures for algorithm performance. Additionally, to reduce the effects of randomness, 4-fold stratified cross validation is used to determine algorithm performance and repeated three times.

The remainder of this paper is organized as follows. Section 2 reviews previous related works focusing on the advantages and limitations of existing algorithms. Section 3 describes the proposed NI-MWMOTE method in details. Section 4 describes the experimental design and shows the results. Conclusions and future works are given in Section 5.

2. Related works

Lots of works have already been done for imbalanced classification problems. These works are mainly concentrated on three levels: data preprocessing, feature selection and classification algorithm improvement (Cao et al., 2017; Diez-Pastor, Rodriguez, Garcia-Osorio, & Kuncheva, 2015; Fan, Wang, Li, Gao, & Zha, 2017; Galar, Fernandez, Barrenechea, Bustince, & Herrera, 2012; Guo et al., 2017). Preprocessing (resampling) techniques increase (or decrease) the number of minority instances (or majority instances) by certain criteria to alleviate imbalances. Feature selection methods improve the recognition rate of minority instances by selecting features with better distinguishing performance. Classification algorithm improvement methods increase the weights on minority instances by improving the flaws in standard classifiers. While there is no single best way to address all the imbalanced classification problems, in recent years, resampling techniques have shown great potential because of their higher versatility (Barua et al., 2014; Nekooimehr & Lai-Yuen, 2016). Thus, in this section, we only provide a brief overview of the resampling techniques of interest, and details of the other works can be found in (Galar et al., 2012; Guo et al., 2017; He & Garcia, 2009).

While resampling techniques cannot simulate real instance distributions, classifiers can learn more from the artificial balanced datasets than the original imbalanced datasets (Barua et al., 2014; Chawla, Cieslak, Hall, & Joshi, 2008). These techniques generally have two major categories. Undersampling techniques (Vuttipittayamongkol & Elyan, 2020; Vuttipittayamongkol, Elyan, Petrovski, & Jayne, 2018) balance the datasets by deleting majority instances, while oversampling techniques balance the dataset by copying or synthesizing minority instances. However, undersampling techniques are less efficient than oversampling techniques. This is because they are easy to remove instances with important information in majority classes, especially when the data is rare (He et al., 2008; Nekooimehr & Lai-Yuen, 2016; Zhou, 2013).

Therefore, oversampling techniques are more favored by researchers.

Random oversampling (ROS) randomly copies minority instances until the datasets are balanced. Since its implementation process is extremely simple, it may be the most commonly used method by practitioners (Douzas et al., 2018). However, it may be overly focused on certain instance or even noise, which tends to cause over-fitting (Batista, Prati, & Monard, 2004; Chawla et al., 2002).

To overcome the risk of over-fitting of ROS, Chawla et al. (2002) proposed the synthetic minority over-sampling technique (SMOTE), which balances the datasets by artificial synthesis rather than random replication. As shown in Fig. 1(a), SMOTE first randomly selects a minority instance x , and then randomly selects a candidate instance y among the k nearest neighbors of minority class. Finally, a new instance g is synthesized by linear interpolation according to $g = x + (y - x) \times \alpha$, where $\alpha \in [0, 1]$ is a random weight.

However, this may lead to over-generalization of SMOTE (Lopez, Fernandez, Garcia, Palade, & Herrera, 2013). This is because it does not consider the actual distribution of the datasets, thereby increasing the overlap different classes (Nekooimehr & Lai-Yuen, 2016). This can be caused by two reasons. 1) It is susceptible to noise. As shown in Fig. 1(b), the sampled instance is a noisy point located in the majority class. Assuming $k = 4$, it is not difficult to find that the newly synthesized instances are also included in the majority class, which adds to the overlap between the categories. 2) In the absence of prior experience, it is difficult to select a reasonable k value. As shown in Fig. 1(c), the sampled instance is located in a small cluster with sparse data. Assuming $k = 4$, it is not difficult to find that newly synthesized samples may be included in the majority class, which also exacerbates the overlap between categories. Additionally, SMOTE does not address within-class imbalances and small separation problems. This is because after oversampling, the data-intensive minority areas may still be dense, while the data-sparse minority areas may remain sparse (Douzas et al., 2018; Prati, Batista, & Monard, 2004).

Therefore, researchers have made lots of modifications and extensions to SMOTE. These modifications usually resolve some of the flaws of the original methods, such as avoiding noise effects and over-generalization or eliminating within-class imbalances (Douzas et al., 2018; Vanhoeyveld & Martens, 2018).

2.1. In terms of avoiding noise effects and over-generalization

BSMOTE (Han et al., 2005) refines oversampling rules based on the distribution of the k nearest neighbors of minority instances, which makes new instances only synthesize at dangerous

instances (or boundary instances). While BSMOTE can avoid the influence of the noise shown in Fig. 1(b), it may still not avoid the overlapping instances shown in Fig. 1(c). This is also caused by a reasonable k value that cannot be determined in advance. ADASYN (He et al., 2008) assigns weights based on the distribution of the k nearest neighbors of minority instances, so that the minority instances with more majority samples in their neighbors have a higher probability of being sampled. However, both ADASYN and BSMOTE may not find all the minority instances close to the decision boundary (Barua et al., 2014; Nekooimehr & Lai-Yuen, 2016). Safe-level-SMOTE (Bunkhumpornpat et al., 2009) calculates the “safe-level” values of all the minority instances and then synthesizes instances closer to the largest safe level value. However, Safe-level-SMOTE may cause over-fitting because new synthetic instances are forced to be generated farther from the decision boundary (Nekooimehr & Lai-Yuen, 2016). Additionally, it must be mentioned that in recent years, to address the imbalanced classification problems not modelled by feature vectors (e.g. time series classification problems (Cao, Li, Woon, & Ng, 2013; Gong & Chen, 2016; Marteau & Gibet, 2015), signature verification problems (Fischer & Plamondon, 2017) and music genre recognition problems (Castellanos, Valero-Mas, Calvo-Zaragoza, & Rico-Juan, 2018), etc.), researchers have further expanded the above SMOTE-based algorithms. Cao et al. (2013) proposed an integrated oversampling (INOS) technique for imbalanced time series classification problems, which mainly uses the enhanced structure preserving oversampling (ESPO) (Cao et al., 2013) to estimate the covariance of minority instances to guide the generation of the most minority instances. Then, it draws on the ideas of ADASYN to guide the generation of the remaining minority instances. INOS aims to gain a balanced dataset that is as realistic as possible. However, ESPO’s covariance idea relies heavily on the distribution of the original minority instances (Gong & Chen, 2016). Further, based on the ADASYN idea, it may be not possible to effectively handle important boundary instances. All of these will cause INOS to be susceptible to noise, thus limiting the practicality of INOS. Similarly, to address more complex imbalanced sequence classification problems (e.g. highly imbalanced, noisy, varying length and multi-variate sequential data, etc.), Gong et al. (Gong & Chen, 2016) proposed the model-based oversampling (MBOS) technique. MBOS uses a recurrent neural network to learn the generative mechanics for sequences as representations for the corresponding sequences. Then, it utilizes these generative models to form a kernel to capture the similarity between different sequences. Finally, SMOTE is performed in the kernel feature space to generate instances. While MBOS considers the mahalanobis distance metric in kernel learning to ensure that different classes of sequences maintain spacing, it is assumed that the sequences are linearly separable.

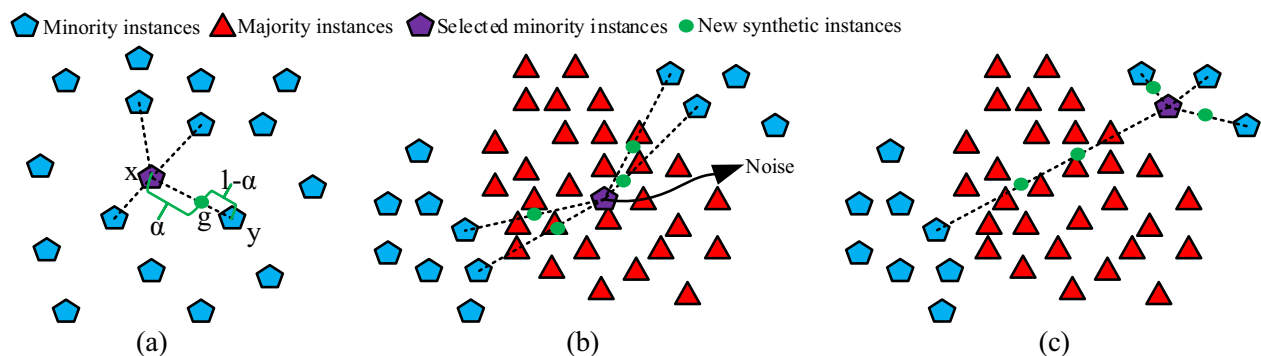


Fig. 1. (a) SMOTE linearly interpolates a randomly selected minority instance using k -nn method ($k = 4$). (b) SMOTE may generate more minority instances in majority regions closer to the noise ($k = 4$), which will make the imbalance classification more difficult. (c) SMOTE may generate some overlapping instances when the minority class has multiple sub-clusters and the k -value is unreasonable ($k = 4$).

Unfortunately, not all sequences are separable in the kernel space, that is, there may be overlapping areas, so performing SMOTE in the kernel space at this time will be susceptible to noise. Therefore, MBOS is not completely immune to noise, which is also the reason why its robustness is not significantly stronger than other techniques (Gong & Chen, 2016). Recently, to solve the imbalanced classification problems such as music genre recognition and signature verification, Castellanos et al. (2018) proposed an oversampling technique suitable for character space, which encodes the instances as strings, and then performs SMOTE (B1 or B2) in the string space according to the Edit Distance instead of the Euclidean Distance to balance the datasets. While the algorithm has achieved certain success, it inherits the shortcomings of SMOTE and others, that is, it cannot effectively process the noise in the character space. While the above algorithms overcome between-class imbalances to some extent, they still do not address within-class imbalances.

2.2. In terms of eliminating within-class imbalances

In this regard, the researchers first decompose the imbalanced datasets into some small sub-clusters and then use sampling methods to change the size of the sub-clusters. Cluster-SMOTE (Cieslak et al., 2006) uses K-means algorithm to divide the minority class into multiple sub-clusters, and then performs SMOTE in each sub-cluster. The stated goal of Cluster-SMOTE is to boost class regions by creating instances within naturally occurring clusters of the minority class (Douzas et al., 2018). While the method may alleviate between-class imbalances, it does not help to avoid over-generalization and eliminate the small separation sub-cluster problems (Douzas et al., 2018; Nekooimehr & Lai-Yuen, 2016) and avoid the effects of noise. DBSMOTE (Bunkhumpornpat et al., 2012) performs DBSCAN clustering to discover the clusters of arbitrary shape, then performs SMOTE within each sub-cluster and makes the new synthetic samples closer to the sub-cluster's pseudo centroids. While the algorithm avoids the appearance of synthetic instances in majority classes, these instances are dense near the pseudo centroid and are sparse away from the pseudo centroid. That is, it ignores the importance of the approaching decision boundary instance, which may lead to over-fitting. CURE-SMOTE (Ma & Fan, 2017) first clusters minority instances using CURE (Clustering using representatives) to remove noise and outlines from the original samples, and then performs SMOTE between the representative point and the center point. Similar to DBSMOTE, while CURE-SMOTE can avoid new instances from appearing in majority classes to some extent, these instances are dense near the center point and sparse away from the center point. Therefore, CURE-SMOTE also easily leads to over-fitting. Recently, Douzas et al. (2018) proposed a heuristic oversampling method based on K-means and SMOTE, which first clusters the overall imbalanced datasets using K-means instead of only minority classes, and then filters the clusters with more majority instances than minority instances. Next, the respective sampling weights are calculated based on the minority class density in the remaining sub-clusters, and finally each sub-cluster is oversampled using SMOTE. Similarly, although K-means SMOTE avoids noise interference problems and synthesizes a large number of relatively safer samples, it is easy to ignore the importance of instances closer to decision boundaries and small minority sub-clusters located in majority class. Now let us give an example. As shown in Fig. 2(a), the imbalanced dataset is divided into three sub-clusters, and the IR1-IR3 values are 8/40, 7/10, 15/3, respectively. However, according to the rules of K-means SMOTE, only when $IR > 1$, the sub-cluster will not be filtered out, so Cluster1 and Cluster 2 will be ignored, which is obviously unreasonable. This is because Cluster 4 in Cluster 1 is an important small sub-

cluster of minority class and should not be ignored. The minority class boundary instances in Cluster 2 are sparse and the majority class boundary instances in Cluster 2 are dense, so they should be valued rather than being ignored. These will lead to the synthesis of too many new instances in Cluster 3, which makes it easier to over-fitting.

In addition to the above improvements and extensions based on SMOTE. In 2014, a new majority weighted minority oversampling technique (MWMOTE) (Barua et al., 2014) was proposed to overcome the shortcomings of the knn based oversampling algorithms. Unlike the knn based algorithms (e.g. BSMOTE and ADASYN, etc.), MWMOTE finds boundary instances in two steps: 1) Calculates the k majority neighbors of each minority instance to obtain the set of majority boundary samples (S_{bmaj}), and 2) Calculate k minority neighbors of each S_{bmaj} to obtain the set of minority boundary samples (S_{imin}). Then, MWMOTE assigns the sampled weights to every minority boundary instances by the Euclidean distance from the majority instances to the minority boundary instances. Finally, MWMOTE generates useful minority instances using hierarchical clustering. While the MWMOTE algorithm strives to overcome within-class imbalances, it is easy to ignore small minority sub-clusters far away from majority instances when identifying minority boundary instances. However, these small sub-clusters are also very important (Nekooimehr & Lai-Yuen, 2016). Now let us give an example. As shown in Fig. 2(b), the minority instances in the dataset have three sub-clusters Cluster 1- Cluster 3, assuming $k_2 = k_3 = 5$, it is not difficult to find that the boundary samples of Cluster 1 and Cluster 3 are found. However, the boundary samples of Cluster 2 are ignored, which is obviously unreasonable. While this problem can be addressed by increasing the k_2 and k_3 values, the number of boundary samples must increase as the value of k increases, which may be fatal to the effectiveness of MWMOTE. This is because it will be difficult to find more useful boundary samples in other sub-clusters. To overcome this shortcoming of MWMOTE, Nekooimehr and Lai-Yuen (2016) proposed an adaptive semi-supervised weighted oversampling (A-SUWO), which uses a semi-supervised hierarchical clustering approach to cluster minority instances before boundary samples' identification. Then, classification complexity and cross-validation are used to adaptively determine the number of new instances that need to be synthesized in each sub-cluster. Compared with MWMOTE, A-SUWO strives to find all boundary samples of minority sub-clusters to improve the performance of the algorithm, while the algorithm is a quite complex technology (Douzas et al., 2018). Especially the semi-supervised hierarchical clustering approach may make the sub-clusters too complicated, which will lead to poor sampling.

Additionally, algorithms (e.g. BSMOTE, ADASYN, MWMOTE and A-SUWO, etc.) have one thing in common, that is, they all used the knn based noise processing method. The principle of this knn method is that if an instance's k nearest neighbors are all the instances of other classes, it will be considered as a noise. However, is the noise processing method reasonable? Now let us give another example. As shown in Fig. 2(c), assuming $k = 5$, it will not be difficult to find that all five neighbors of p1, p2 and p3 are the majority instances. Therefore, all these three points will be considered as noisy points. While p1 and p2 are considered as noisy points without any objection, p3 is a boundary point which contains important information, therefore, it is unreasonable to regard it as a noisy point. While p3 can be avoided as a noisy point by increasing the value of k, p2 will probably not be considered as a noisy point at this time, which is obviously unreasonable. Therefore, knn method has serious deficiencies in identifying noise.

In summary, many recent works have been proposed for the improvement of oversampling methods, where most of the methods use clustering techniques before applying SMOTE or

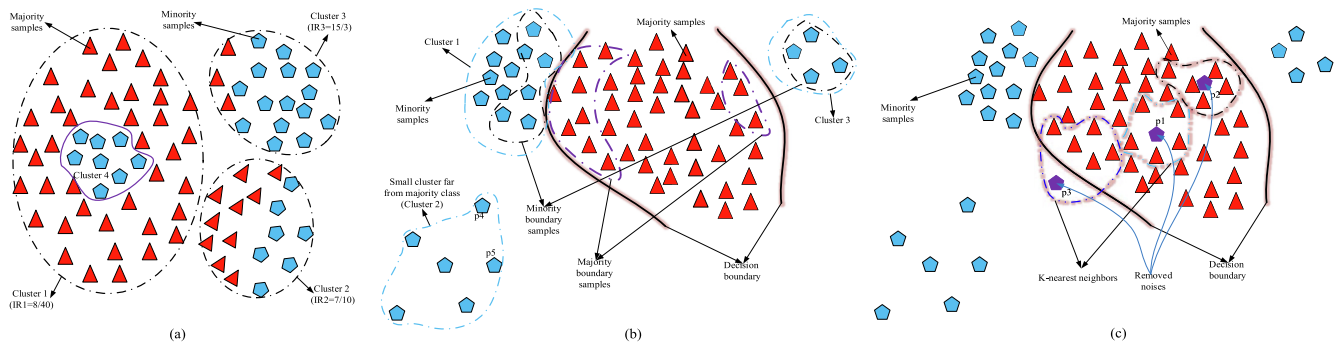


Fig. 2. (a) Shortcomings of k-means SMOTE, easily ignore small clusters and boundary samples. (b) MWMOTE is easy to ignore small clusters that is far away from majority instances ($k1 = 4$). (c) Identification of noise points using k-nn method ($k = 4$).

MWMOTE. While most of them have managed to overcome some of the weaknesses of existing oversampling algorithms, none of them can accurately handle noise and overcome both between-class imbalances and within-class imbalances at the same time. Additionally, some sampling techniques achieve their respective improvements at the expense of high complexity, making them difficult to implement and use, especially when the amount of dataset is larger and the complexity of dataset is higher.

3. Noise-immunity majority weighted minority oversampling Technique (NI-MWMOTE)

NI-MWMOTE consists of three main steps: 1) Adaptive Noise Removing, 2) Sub-clusters Determining and Adaptive Sub-cluster Sizing, and 3) Synthetic Instance Generation using MWMOTE.

The first step mainly eliminates true noisy points. Specifically, a method of adaptive processing noise by iterative and misclassification error was proposed for the first time. Different from the traditional noise processing methods based on knn, based on Euclidean distance, based on Euclidean distance combined with k-nearest neighbor density, and based on clustering methods, NI-MWMOTE's method of processing noise is more intelligent and effective.

In the second step, NI-MWMOTE uses unsupervised AHC to cluster minority instances, which mitigates the possibility of ignoring small sub-clusters. Then, an adaptive sub-cluster sizing method (Nekooimehr & Lai-Yuen, 2016) was used to determine the oversampling size of each minority sub-cluster, which can well address within-class imbalanced problems.

In the final step, we use MWMOTE to synthesize new instances in each minority sub-cluster. The MWMOTE algorithm in this step is different from the standard MWMOTE (Barua et al., 2014), which does not apply knn based noise processing method.

3.1. Adaptive noise removing

3.1.1. Motivations of adaptive noise removing

From the previous discussion, we already know that the knn method only considers the distribution of k nearest neighbors to remove noise. Therefore, it exhibits a certain blindness, i.e., it is easy to treat the boundary points containing important information as noise or treat the true noisy points as normal instances. Besides the knn method of sampling techniques, Batuwita and Lin et al. (Batuwita & Palade, 2010; Lin & Wang, 2002) utilized a noise recognition method based on the Euclidean distance of sample-to-class-center for FSVM-CIL (Fuzzy Support Vector Machines for Class Imbalance Learning) and FSVM (Fuzzy Support Vector Machines) classifiers. However, this method is not applicable to non-spherical datasets. To end this, Ju et al. (Ju, Cao, & Gu, 2016) considered both the above Euclidean distance and the samples' k

nearest neighbor densities. While the noise recognition probability is improved using this method, it cannot accurately identify the number of noisy points. Therefore, it is imminent to propose an effective noise processing method. First, we need to clarify 2 issues:

1) So what are the characteristics of noisy points?

For easy explanation, we will use the cluster 2 in Fig. 2 (c) as an example to analyse the differences between true noise (p1) and pseudo (false) noise (p3). Fig. 3(a) shows a comparison of the distances of the suspected noise points p1 and p3 to the majority (minority) class center, where $d1$ and $d2$ represent the Euclidean distances between the two points and the minority class center, and $d3$ and $d4$ represent the Euclidean distances between the two points and the majority class center. Analysis of Fig. 3(a) will not difficult to find that $d1 < d2$, $d3 > d4$, which indicates that the distance of true noisy point from the minority class center is greater than that of the false noisy point. Conversely, the distance from the majority class center is less than that of the false noisy point. That is, a suspected noisy point is farther away from the minority class center, and the closer it is to the majority class center, the easier it is to be a true noise. Fig. 3(b) shows a comparison of the neighbour densities of the suspected noise points p1 and p3 based on the majority (minority) class (assuming the neighbour density parameter $k' = 5$), where $den1$ and $den2$ represent the neighbourhood densities of the two points in the minority class, and $den3$ and $den4$ represent their neighbourhood densities in the majority class. Analysis of Fig. 3(b) will not difficult to find that $den1 < den2$, $den3 > den4$, which indicates that the nearest neighbourhood density of the true noisy point in the majority class is greater than that of the false noisy point. Conversely, in the minority class, its density in the neighbourhood is less than that of the false noisy point. That is, the density of a suspected point in majority class is larger, and the smaller the density of it in minority class, the easier it is considered as a true noise.

2) What the relationship of the misclassification error of original imbalanced datasets and noise?

To explore the relationship between the two, we intend to investigate by sampling the original imbalanced data. *Settings for the sampling algorithm:* We try to select a simple and susceptible noise-affected algorithm. That is, the selected synthetic oversampling method should be easily over-generalized, and its parameter selection does not need to be quite accurate. Therefore, we selected SMOTE ($k = 5$ in this paper) for the experiments. The reason for not choosing other methods is that while the ROS algorithm is very simple, it is prone to over-fitting. While, more complex algorithms (e.g. BSMOTE, MWMOTE and A-SUWO, etc.) have their own noise processing algorithms. *Settings for the testing sets and evaluation*

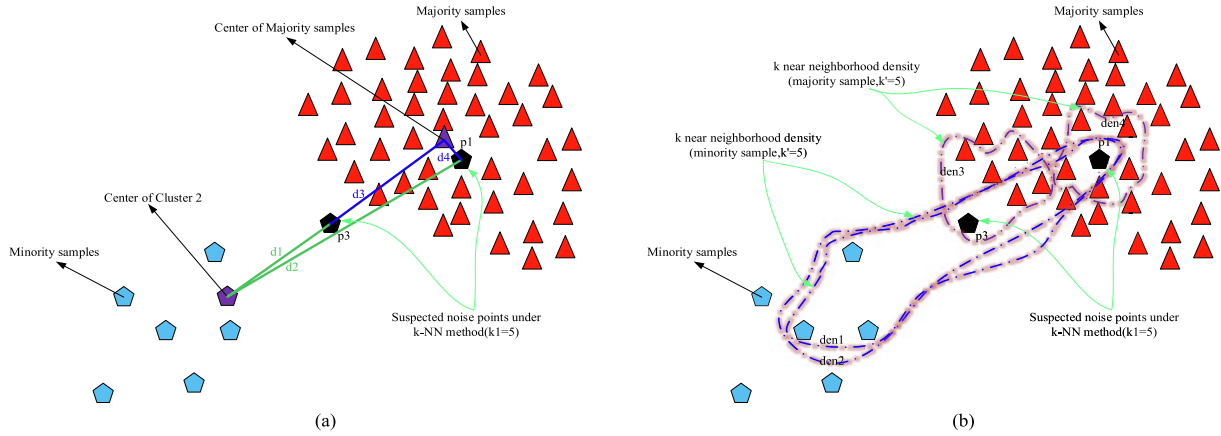


Fig. 3. (a) Comparison the sample distances of suspected the noise points (b) Comparison the k' (assuming $k' = 5$) nearest neighbor densities of the suspected noise points.

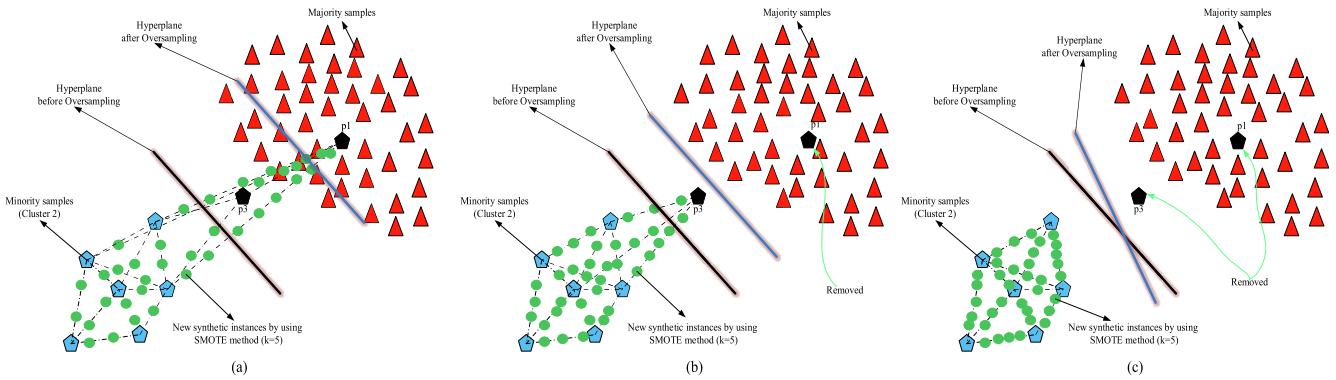


Fig. 4. Relationship between noises processing number and classification hyperplane using SMOTE ($k = 5$), (a) None noises are removed, (b) First one noise is removed, (c) First two noises are removed.

measure: We selected the sub-cluster being processed and all the majority instances as the testing sets. And the G-mean (see Section 4 for details) is selected as the final evaluation measure.

Similarly, we still use the Cluster 2 as an example. Where both $p1$ and $p3$ are suspicious noisy points, and $p1$ is more likely to be true noise than $p3$. Fig. 4(a)–(c) show the relationship between the decision boundary and the sampled datasets when the number of noise deletions is 0–2. It is not difficult to observe that the three testing sets reveal three different misclassification errors ($error_a > error_c > error_b$). Therefore, it can be easily concluded that the reasonable degree of noise processing can be reflected by the misclassification error on the original dataset. That is, the more reasonable the noise processing place, the smaller the misclassification error of the original dataset. Further, the G-mean values satisfy $G - mean_b > G - mean_c > G - mean_a$. That is, the misclassification error and the G-mean values are just inversely related, i.e., the larger the G-mean value is, the smaller the error is. Therefore, the misclassification errors can be expressed by the G-mean values.

3.1.2. Method for calculating the probability that suspicious noise is true noise

Through the analysis of 1) in Section 3.1.1, we have known the differences between real noise and suspicious noise. Therefore, we define the method for calculating the probability that suspicious noise is true noise as follows.

1. Calculate the distance from the suspected noise point to the majority(minority) class center

First, assuming $x_i^p, i = (1, 2, 3, \dots, m)$ and $x_i^n, i = (1, 2, 3, \dots, M)$ represent the minority and majority class sample sets, respectively, so we compute the corresponding of the majority (minority) class centers ($x^{p_{Center}}, x^{N_{Center}}$) using the following equations.

$$x^{p_{Center}} = \sum_{i=1}^m \frac{x_i^p}{m} \quad (1)$$

$$x^{N_{Center}} = \sum_{i=1}^M \frac{x_i^n}{M} \quad (2)$$

Then, assuming $x_j^{s-noise}, j = (1, 2, 3, \dots, N_s - noise)$ represents the dataset of the suspected noise points, we compute the Euclidean distances from the suspected samples to the majority (minority) class centers ($d_i^{p_{Center}}, d_i^{N_{Center}}$) using the following equations.

$$d_i^{p_{Center}} = \|x_j^{s-noise} - x^{p_{Center}}\|; i = 1, 2, 3, \dots, N_s - noise \quad (3)$$

$$d_i^{N_{Center}} = \|x_j^{s-noise} - x^{N_{Center}}\|; i = 1, 2, 3, \dots, N_s - noise \quad (4)$$

Finally, to eliminate the dimensional influence on the two distances determined above, we need to standardise them. The normalising processing of $d_i^{p_{Center}}, d_i^{N_{Center}}$ is as follows.

$$d_i^{p_{Center}(0.5,1)} = \frac{d_i^{p_{Center}} - \min(d_i^{p_{Center}})}{\max(d_i^{p_{Center}}) - \min(d_i^{p_{Center}}) + \delta}; \quad i = 1, 2, 3, \dots, N_s - noise \quad (5)$$

$$d_i^{N_{Center(0.5,1)}} = \frac{d_i^{N_{Center}} - \min(d_i^{N_{Center}})}{\max(d_i^{N_{Center}}) - \min(d_i^{N_{Center}}) + \delta};$$

$$i = 1, 2, 3, \dots, N_s - \text{noise} \quad (6)$$

where $d_i^{P_{Center(0.5,1)}}$ and $d_i^{N_{Center(0.5,1)}}$ are the normalisations of $d_i^{P_{Center}}$ and $d_i^{N_{Center}}$, respectively. $\delta (\delta = 1.0e^{-5})$ is a small positive number which ensures that the denominators of Eqs. (5) and (6) are meaningful.

2. Calculate the nearest neighbor density of the suspected noise point

Assuming $x_k^p \in N_{knn}^p(x_j^{s-\text{noise}}), k = 1, 2, 3, \dots, k'$ and $x_k^N \in N_{knn}^N(x_j^{s-\text{noise}}), k = 1, 2, 3, \dots, k'$ represent the samples of k' nearest neighbors of $x_j^{s-\text{noise}}$ in the minority and majority classes, respectively, we compute the k' neighbourhood densities of the suspected noisy points $(D_j^{P_{knn}}, D_j^{N_{knn}})$ using the following equations.

$$D_j^{P_{knn}} = \sum_{k=1}^{k'} \frac{\|x_j^p - x_k^p\|}{k'}; j = 1, 2, 3, \dots, N_s - \text{noise} \quad (7)$$

$$D_j^{N_{knn}} = \sum_{k=1}^{k'} \frac{\|x_j^N - x_k^N\|}{k'}; j = 1, 2, 3, \dots, N_s - \text{noise} \quad (8)$$

Similarly, in order to eliminate the dimensional influence on the two densities determined above, we need to standardise them. The normalising processing of $D_j^{P_{knn}}$ and $D_j^{N_{knn}}$ is as follows.

$$D_j^{P_{knn(0.5,1)}} = \frac{D_j^{P_{knn}} - \min(D_j^{P_{knn}})}{\max(D_j^{P_{knn}}) - \min(D_j^{P_{knn}}) + \delta}; j = 1, 2, 3, \dots, N_s - \text{noise} \quad (9)$$

$$D_j^{N_{knn(0.5,1)}} = \frac{D_j^{N_{knn}} - \min(D_j^{N_{knn}})}{\max(D_j^{N_{knn}}) - \min(D_j^{N_{knn}}) + \delta}; j = 1, 2, 3, \dots, N_s - \text{noise} \quad (10)$$

where $D_j^{P_{knn(0.5,1)}}$ and $D_j^{N_{knn(0.5,1)}}$ are the normalisations of $D_j^{P_{knn}}$ and $D_j^{N_{knn}}$, respectively. δ is the same as above which ensures that the denominators of Eqs. (9) and (10) are meaningful.

3. Calculate the decision set of suspected noise

From the previous analysis, we already know that the suspected noise point is farther away from minority class center, and the closer it is to the majority class center, the easier it is to be considered as true noise. Further, the density of the majority class is larger, and the smaller the density of the minority class, the easier it is to be true noise. Therefore, we define the calculation method of the decision set as follows.

$$N_j^{s-\text{noise}} = 1 - w \times d_j^{P_{Center(0.5,1)}} \times (1 - d_j^{N_{Center(0.5,1)}}) - (1 - w) \times D_j^{P_{knn(0.5,1)}} \times (1 - D_j^{N_{knn(0.5,1)}}); j = 1, 2, 3, \dots, N_s - \text{noise} \quad (11)$$

$$N^{s-\text{noise}} = \text{sort}\{N_j^{s-\text{noise}}\}; j = 1, 2, 3, \dots, N_s - \text{noise} \quad (12)$$

where $N^{s-\text{noise}}$ is the suspected noises decision set, $N_j^{s-\text{noise}}$ is the i -th suspected noise decision value, $w \in \{0, 0.1, 0.2, \dots, 1\}$ is the weight factor which is used to adjust the relative significances of sample spacing and density. Moreover, it is not difficult to find that the smaller the value of $N_j^{s-\text{noise}}$ for a suspected noise, the easier it is to be judged as a noisy point.

3.1.3. Flow of the adaptive noise removing method

Based on Sections 3.1.1 and 3.1.2, the specific steps of our proposed adaptive noise removing method are as follows:

Algorithm 1. Adaptive Noise Removing ($S_{\text{maj}}, S_{\text{min}}, k1, k', w$)

Input:

- S_{maj} : Set of majority instances.
- S_{min} : Set of minority instances.
- $k1$: Number of neighbors used for identifying suspicious noisy minority instances using knn method.
- k' : Number of neighbors used for calculating neighborhood density of majority or minority instances.
- w : Weight between the suspicious noise's Euclidean distance to class center and its class nearest neighbor density.

Procedure Begin:

1. Combine S_{maj} and S_{min} to form a training set, then use the knn method to identify suspicious noise of $S_{\text{min}}, C_{j=1, \dots, n}^{\text{sus-noise}}$ and the S_{maj} (or S_{min}) class center using Eqs. (1)–(6), $d_j^{P_{Center}}, d_j^{N_{Center}}$.
2. Calculate the Euclidean distances between each $C_{j=1, \dots, n}^{\text{sus-noise}}$ and the S_{maj} (S_{min}) class center using Eqs. (7)–(10), $D_j^{P_{knn}}, D_j^{N_{knn}}$.
3. Calculate the neighborhood densities of each $C_{j=1, \dots, n}^{\text{sus-noise}}$ in S_{maj} (S_{min}) using Eqs. (7)–(10), $D_j^{P_{knn}}, D_j^{N_{knn}}$.
4. Calculate the probabilities of each $C_{j=1, \dots, n}^{\text{sus-noise}}$ being true noise are calculated using Eqs. (11) and (12) and sorted, $\text{sort}\{N_{j=1, \dots, n}^{\text{sus-noise}}\}$.
5. According to $\text{sort}\{N_{j=1, \dots, n}^{\text{sus-noise}}\}$, do the followings

Do for j (number of suspected noises removed) $\in \{0, 1, \dots, n\}$

Perform the operation a-c.

a. According to $\text{sort}\{N_{j=1, \dots, n}^{\text{sus-noise}}\}$, delete the suspected noises with the first j values and get S_{min}' .

b. Perform SMOTE oversampling on the dataset $C^{\text{imbalanced}} = [S_{\text{maj}}; S_{\text{min}}']$ and get the balanced dataset C^{balanced} .

c. First, input C^{balanced} as the training set into classifier, then use $C^{\text{imbalanced}}$ as the testing set input classifier, and finally calculate the G - mean value of the test set and store them G_j .

End Loop

Output $S_{\text{min}}'_{\text{best}}$ corresponding to the optimal G_j value.

END

Output: The minority instances after noise processing, $S_{\text{min}}'_{\text{best}}$.

Please note that in the Step 5 of **Algorithm 1**, there are several points that need to be explained.

- 1) To reduce the random impact, it is necessary to perform SMOTE (5 times in this paper) multiple times, and use the final statistical average values as the final decision values of G-mean.
- 2) When selecting a classifier, it needs to be consistent with the classifier that processes the balanced dataset after sampling, and there is no need to optimize its parameters. The reason is that we only focus on the error rate, not its accuracy.
- 3) When there are multiple G-mean maximum values, we need to select the noise processing scheme when the first maximum occurs, which not only preserves the distribution of the original datasets to the maximum, but also retains minority instances with more information.

3.2. Sub-clusters determining and adaptive Sub-cluster sizing

When standard MWMOTE identifies boundary instances, it is easy to ignore minority sub-cluster far away from majority instances (Nekooimehr & Lai-Yuen, 2016), which may be fatal to alleviate within-class imbalances. Therefore, Nekooimehr et al. used a semi-supervised hierarchical clustering approach to prevent small sub-clusters from being ignored before identifying boundary samples, but this clustering method is too complicated. Not only is it not applicable to limited and imbalanced datasets with between-class overlaps, but also the computational cost of datasets with large amounts of data and high spatial structure complexity will be excessive. Therefore, for NI-MWMOTE, we use an unsupervised clustering method to prevent small minority sub-clusters from being ignored. While this comes at the expense of clustering accuracy, its application will be more extensive and easy to implement.

Further, in general, to better solve the within-class imbalanced problems, the number of samples to be synthesized in different sub-clusters should be different. Therefore, existing algorithms (e.g. K-means SMOTE (Douzas et al., 2018)) allocate the number of samples to be synthesized in each sub-cluster according to the density of each sub-cluster to address within-class imbalances. While these methods are simple and easy to implement, there are limitations to determining the number of instances based on density. This is because not all the sub-clusters are evenly distributed, especially when the number of samples in some sub-clusters is very limited. Fortunately, Nekooimehr and Lai-Yuen (2016) proposed a method based on classification complexity and cross-validation to adaptively determine the number of samples to be synthesized in each sub-cluster. Obviously, this method is not limited to the distribution of sub-clusters. Therefore, our proposed NI-MWMOTE will also use standardized mean error rate (classification complexity) and cross-validation to determine the oversampling size of each minority sub-cluster.

In summary, the main steps of Sub-clusters determining and Adaptive sub-cluster sizing in NI-MWMOTE are as follows.

Algorithm 2. Sub-clusters determining and adaptive sub-cluster sizing (S_{maj} , $S_{min}'_{best}$, m , IDX_min , K , N)

Input:

- S_{maj} : Set of majority instances.
- $Cmin'_{i=1,...,m}$: Minority instances after noise processing.
- m : Number of minority class sub-clusters using AHC.
- IDX_min : $Cmin'_{i=1,...,m}$'s sub-cluster labels (1,...,m).
- K : Fold number of cross-validation.
- N : Number of new instances synthesized using NI-MWMOTE.

Procedure Begin:

1. Perform AHC clustering method to divide $S_{min}'_{best}$ into m sub-clusters, $Cmin'_{i=1,...,m}$.
2. Randomly divide each minority sub-cluster belonging to $Cmin'_{i=1,...,m}$ into K similar size partitions.
3. Let $i = 1$, and let $K-1$ partitions from $Cmin'_1$ and all majority instances (S_{maj}) as the training set, and the remaining one partition as the test set. Then, SMOTE oversampling is performed on the training set, and training is performed using a linear classifier (see (Nekooimehr & Lai-Yuen, 2016) for reasons).
4. Calculate the error rate of the above testing set using $\varepsilon_k = \varepsilon_k / N'$, where ε_k is determined as the number of minority instances misclassified as majority for each $Cmin'_i$ in fold k , and N' is the number of instances of corresponding minority sub-cluster.
5. Select $Cmin'_i$'s other $K-1$ minority partitions and S_{maj} as the next training set, and the remaining one partition as the next testing set, that is, repeat steps. 2–4 until the maximum number of cycles K is reached.
6. Calculate the average error rate of $Cmin'_i$ using $\bar{\varepsilon} = \sum_{k=1}^K \varepsilon_k / K$. Then, Let $i = i + 1$, repeat steps. 2–5 until each $Cmin'_{i=1,...,m}$ is selected and performs the above operations.
7. Standardize average error rate using $\bar{\varepsilon}_i^* = \bar{\varepsilon}_i / \sum_{i=1}^m \bar{\varepsilon}_i$.
8. Determine the number of minority instances that need to be synthesized in each $Cmin'_{i=1,...,m}$, using $n_i^* = N * \bar{\varepsilon}_i^* - n_{sub-cluster i}$, $i = 1, \dots, m$, where $n_{sub-cluster i}$ is the number of minority instances in $Cmin'_i$.

END

Output: The number of samples needed to be synthesized for each minority sub-cluster, $n_i^* = 1, \dots, m$.

Please note for Algorithm 2, there are several points that need to be explained.

- 1) Why use AHC in the Step 1 of Algorithm 2? Comparing with AHC (S. B. Zhou, Xu, & Liu, 2017), K-means method (Douzas et al., 2018) is sensitive to initial values and susceptible to noise. DBSCAN (C. Bunkhumpornpat et al., 2012) are not suitable for datasets with uneven density and large cluster spacing, and adjustment of their parameters is complicated. Therefore, we choose AHC to cluster minority instances.
- 2) Algorithm 2 is executed only when the number of minority class sub-clusters in Algorithm 1 is $m > 1$, otherwise the **Algorithm 3** is directly executed.

- 3) The SMOTE in the Step 3 of Algorithm 2 is only used to calculate the misclassification error rate of minority instances, so its parameter k do not need to be setted to a very well range (here, $k = 5$). Additionally, the SMOTE also needs to be executed multiple times to avoid the influence of random error.

3.3. Synthetic instance generation using MWMOTE

In the third step of NI-MWMOTE, we use MWMOTE to synthesize new instances for each minority sub-cluster. The MWMOTE algorithm in this step is different from the standard MWMOTE algorithm in (Barua et al., 2014), which does not apply knn based noise processing method. Therefore, the main steps of MWMOTE in NI-MWMOTE are as follows.

- b) Further, MWMOTE makes $k3 = |S_{\min}|/i$, $i = 1, 2, \dots, n$ ($n \leq |S_{\min}|$). It is not difficult to find that the value of $k3$ decreases by multiples and changes greatly, which is not conducive to finding the optimal recommendation range.
- c) Even if a smaller value of $k3$ cannot find all valid boundary instances, however, according to the instance generation method of MWMOTE (i.e., Step 7 in Algorithm 3), the remaining valid boundary instances still have the opportunities to be sampled because the candidate instances are all the entire individuals in Clusters. Additionally, from the exploration (Barua et al., 2014) of the value range of the parameter $k3$, the AUC (please see Section 4 for details) results of the three imbalanced datasets show that the results under the optimal parameters are

Algorithm 3. MWMOTE used by NI-MWMOTE ($S_{maj}, C_{min}'_i = 1, \dots, m, n_i^* = 1, \dots, m, k2$)

Input:

- S_{maj} : Set of majority instances.
- $C_{min}'_i = 1, \dots, m$: Minority instances after noise processing.
- $n_i^* = 1, \dots, m$: Number of samples needed to be synthesized for each minority sub-cluster.
- $k2$: Number of majority (and minority) neighbors used for constructing informative minority set.

Procedure Begin:

1. Let $i = 1$, Calculate the nearest majority set ($N_{maj}(x_j)$) of each $x_j \in C_{min}'_i$, $j = 1, \dots, n_i^*$ in all major classes, $N_{maj}(x_j)$ consists of the nearest $k2$ majority samples from x_j according to Euclidean distance. Then, combine all the $N_{maj}(x_j)$ to get the majority class boundary sample set, $S_{bmaj} = \bigcup_{x_j \in C_{min}'_i} N_{maj}(x_j)$.
2. Calculate the nearest minority set ($N_{min}(y_j)$) of each $y_j \in S_{bmaj}$ in $C_{min}'_i$, $N_{min}(y_j)$ consists of the nearest $k2$ majority samples from y_j according to Euclidean distance. Then, combine all the $N_{min}(y_j)$ to get the minority class boundary sample set, $S_{imin} = \bigcup_{y_j \in S_{bmaj}} N_{min}(y_j)$.
3. For each $y_j \in S_{bmaj}$ and $x_j \in S_{imin}$, compute the information weight, $I_w(y_j, x_j) = C_f(y_j, x_j) \times D_f(y_j, x_j)$. The specific solution method of $C_f(y_j, x_j)$ and $D_f(y_j, x_j)$ can be found in (Barua et al., 2014).
4. For each $x_j \in S_{imin}$, compute the selection weight $S_w(x_j)$ using $S_w(x_j) = \sum_{y_j \in S_{bmaj}} I_w(y_j, x_j)$.
5. Convert each $S_w(x_j)$ into selection probability $S_p(x_j)$ according to $S_p(x_j) = S_w(x_j) / \sum_{Z_j \in S_{imin}} S_w(Z_j)$.
6. Find the clusters of $C_{min}'_i$. Let, M clusters are formed which are $L_1, \dots, L_M$.
7. Do the following to synthesize new minority instances
Initialize the set, $S_{omin} = C_{min}'_i$
Do for $i = 1, \dots, n_i^*$
 Perform the operation a-d.
 a. Select a sample x from S_{imin} according to probability distribution $\{S_p(x_j)\}$. And x is a member of L_k , $k = 1, \dots, M$.
 b. Select an another sample y , randomly, from cluster L_k .
 c. Synthetic a new instance, s , using $s = x + \alpha \times (y - x)$, where α is a random number in $[0, 1]$.
 d. Add the new instance, s , to S_{omin} , $S_{omin} = S_{omin} \cup \{s\}$
End Loop
 Store the data set after oversampling, $S_1 = S_{omin}$.
8. Let $i = i + 1$, and repeat steps.1-8 until the sampling of m sub-clusters is completed. Finally, S_{maj} and $S_{min} - new = \{S_1, S_2, \dots, S_m\}$ are combined to obtain a balanced dataset, $S_{balanced} = [S_{maj}; S_{min} - new]$.

END

Output: The balanced dataset sampled by NI-MWMOTE, $S_{balanced}$.

Please note for the **Algorithm 3**, there are several points that need to be explained.

- 1) In addition to noise processing, unlike the standard MWMOTE (Barua et al., 2014), the MWMOTE used by NI-MWMOTE only involves 4 parameters, namely $k2$, C_p , C_{fth} and C_{MAX} . Obviously, to simplify the algorithm, we have made $k2$ and $k3$ as the same value, that is, they both use the $k2$ values recommended by (Barua et al., 2014) (i.e., $k2 = 5, 10$ or 20). The reasons are as follows:
 - a) $k3$ is used to determine the number of boundary instances to be sampled. It seems that a large value of $k3$ can make a sufficient number of boundary instances to be sampled. However, the instances generated in this way are over-generalized, which weakens the importance of boundary samples.

slightly different from those under the worst parameters (i.e., 1.08%, 2.73% and 0.66%, respectively), which also confirmed our conjecture. Therefore, we consider $k2$ and $k3$ as the same value.

- 2) For the parameter settings of C_p (control the number of clusters) and C_{fth} & C_{MAX} (smooth and rescal the values of different scaling factors used in weighting the minority class samples), we use the recommended parameters in (Barua et al., 2014), $C_p = 3$, $C_{fth} = 5$, and $C_{MAX} = 2$. Where, C_p chose only one value because the influence of different C_p on the value AUC is minimal (i.e., 1.23%, 1.07% and 0.95%, respectively).

Combining the three algorithms in Sections 3.1 to 3.3, the flow of our proposed NI-MWMOTE is as follows (See Fig. 5):

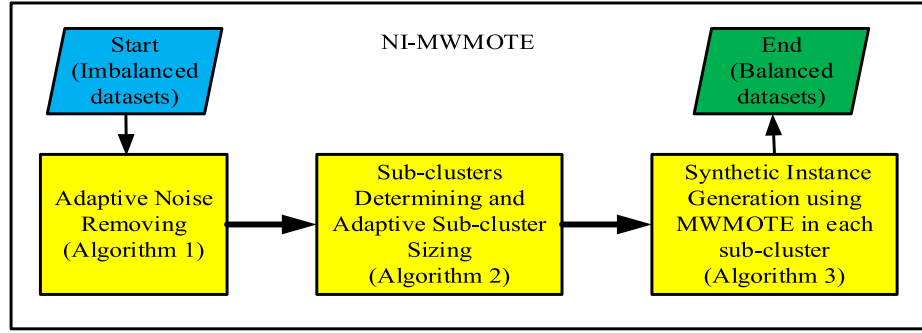


Fig. 5. Flow of NI-MWMOTE.

3.4. Choosing appropriate parameters for NI-MWMOTE

NI-MWMOTE requires 9 parameters (i.e., k_1 , k' , k_2 , C_p , C_{fth} , w , m and K) to choose, where the parameters of k_2 , C_p , C_{fth} and C_{MAX} have been briefly described in Section 3.3. Therefore, avoiding redundancy, we just select five parameters k_1 , k' , w , m and K to explore. Then, in this Section, we will use 6 different imbalanced datasets as examples to briefly explain how to choose appropriate values for these five parameters. Sensitivity analysis of G-mean is also performed by running NI-MWMOTE with different set of values for each parameter. Finally, the results are presented in Fig. 6.

- 1) **k_1** : This parameter is used to determine the number of suspected noise (knn based method), which directly affects the number of subsequent iterations. A smaller k_1 value will result in many minority instances being considered as noise, while a larger k_1 value will result in fewer minority instances being considered as noise. Correspondingly, more suspicious noise will lead to too many iterations, which

increases the computational cost of the algorithm, and this cost may be multiplied. While less suspicious noise can reduce the number of iterations and reduce the computational cost, its processing of noise may be useless. As shown in Fig. 6, while the optimal k_1 values are different for different datasets, it can be determined that the k_1 value should not be too small, and the optimal values are mostly between 3 and 6.

- 2) **k'** : This parameter is used to determine the nearest neighbor density of suspected noise according to majority instances and minority instances. A smaller k' value will cause NI-MWMOTE to be susceptible to noise when calculating the neighbor density. While increasing the k' value can avoid noise interference, an excessive k' value is also unnecessary. This is because an excessive k' value increases the computational cost, and it may not increase the difference between true noise and pseudo noise. Even, when the m value is unreasonably selected and there are ignored small sub-clusters in the same category, this situation will more serious. As shown in Fig. 6, although the optimal k' values are

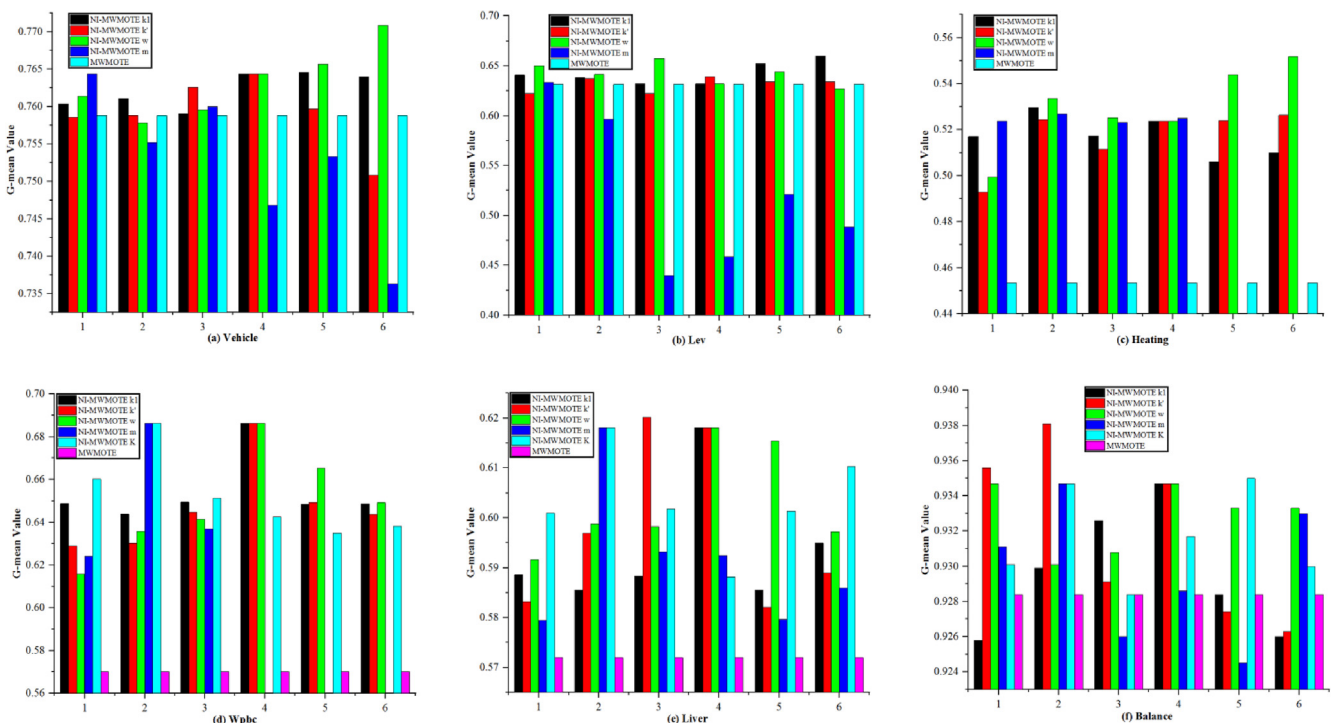


Fig. 6. Sensitivity analysis on NI-MWMOTE parameters using SVM, where the abscissas 1 ~ 6 of k_1 , k' , and K represent 2, 3, ..., 7, the abscissas 1 ~ 6 of w represent 0.2, 0.3, ..., 0.7, and the abscissas 1 ~ 6 of m represent 1, 2, ..., 6.

different for different datasets, it can be determined that the k' value should not be too small or too large, and the optimal values are mostly between **3 and 6**.

- 3) **w**: This parameter is used to adjust the weight between the Euclidean distance from suspected noise to class center and its neighbor density. If $w > 0.5$, the weight of the Euclidean distance from the suspected noise to class center is greater than that of its neighbor density, that is, the Euclidean distance to class center is the main influencing factor. On the contrary, if $w < 0.5$, the weight of the Euclidean distance from the suspected noise to class center is less than that of its neighbor density, that is, the neighbor density is the main influencing factor. Additionally, if $w = 0.5$, the effect of the two is the same. As shown in Fig. 6, although the optimal w values are different for different datasets, it can be determined that w should not be too large or too small, and the optimal values are mostly between **0.3 and 0.6**. This is because a too large or too small w value will inevitably reduce the correction of another influencing factor, especially when a data set is irregularly distributed.
- 4) **m**: This parameter is used to identify small minority sub-clusters far from majority classes to compensate for the blind spots of standard MWMOTE in boundary instances identification. Therefore, the lower limit of m is 1, because such datasets may not have small clusters that are ignored. When m is too large, too many sub-clusters not only increase the computational cost of subsequent sub-cluster size allocation, but also easily lead to over-fitting, which in turn may cause the sampling effect to drop sharply. As shown in Fig. 6, although the optimal m values are different for different datasets, m should not be too large, and the optimal values are mostly between **1 and 3**.
- 5) **K**: This parameter uses to determine the number of folds in NI-MWMOTE's adaptive sub-cluster sizing. It should be noted that when $m = 1$, K does not have any meaning, so K only makes sense when $m > 1$. For the NI-MWMOTE algorithm, the larger the K value, the higher the computational cost. This is because the larger K , the more time it needs to run when determining the misclassification rate for each minority sub-cluster. As shown in Fig. 6, although the optimal K values are different for different datasets, K should not be too large, and the optimal values are mostly between **3 and 6**.

Additionally, it is not difficult to find that in most datasets, although the parameters k_1 , k' , w , m , and K are not optimal, the effect of NI-MWMOTE is still superior to that of standard MWMOTE. In other words, the improved performance of NI-MWMOTE compared to standard MWMOTE is less sensitive to the setting of new parameters.

4. Results and discussion

To verify the effectiveness and superiority of our proposed NI-MWMOTE algorithm, we will evaluate it on 16 publicly available datasets and compare with eight other popular oversampling methods: **1**) Random Oversampling (ROS), **2**) SMOTE (Nitesh V. Chawla et al., 2002), **3**) BSMOTE (H. Han et al., 2005), **4**) ADASYN (He et al., 2008), **5**) Safe-level-SMOTE (Chumphol Bunkhumpornpat et al., 2009), **6**) Cluster-SMOTE (Cieslak et al., 2006), **7**) A-SUWO (Nekooimehr & Lai-Yuen, 2016), and **8**) MWMOTE (Barua et al., 2014). All the programs of these algorithms involved in this paper were implemented using Matlab R2012a on a workstation with 64-bit Operating System, 16.00 GB RAM, and 3.60 GHz CPU.

4.1. Evaluation measures for imbalanced classification

For imbalanced classification, traditional evaluation method based on overall accuracy will no longer be applicable. This is because even if all minority instances were incorrectly classified, a very high overall accuracy can be achieved (Nekooimehr & Lai-Yuen, 2016). Thus, traditional evaluation method will not effectively measure the feasibility of the sampling algorithm. Therefore, in our study, we apply four suitable evaluation measures to compare the different methods: Sensitivity, G-mean, F-measure and Area under Receiving Operator Characteristics Graph (AUC) (Barua et al., 2014; Guo et al., 2017; Nekooimehr & Lai-Yuen, 2016). They can be calculated as follows

$$\text{Sensitivity} = TP / (TP + FN) \quad (13)$$

$$\text{Specificity} = TN / (TN + FP) \quad (14)$$

$$G - \text{mean} = \sqrt{\text{Sensitivity} \times \text{Specificity}} \quad (15)$$

$$PRE = TP / (FP + TP) \quad (16)$$

$$F - \text{measure} = 2 \times \text{Sensitivity} \times PRE / (\text{Sensitivity} + PRE) \quad (17)$$

$$AUC = (\text{Sensitivity} + \text{Specificity}) / 2 \quad (18)$$

where TP is the number of positive (minority) class samples with correct classification, FN is the number of positive class samples with wrong classification, FP is the number of negative (majority) class samples with wrong classification and TN is the number of negative class samples with correct classification. Analysis of Eqs. (13)–(18) reveals that the higher the sensitivity, the higher is the recognition rate of the positive samples. The larger the specificity, the higher is the recognition rate of the negative samples. Only when both the sensitivity and specificity are large is the G-mean value larger and the classification effect better. The F-measure value also works well for the recognition rate of the positive samples. The AUC value can well reflect the accuracy of the model.

4.2. Datasets and parameters setted for imbalanced classification

To ensure the credibility and generality of our results, we selected 24 kinds of realities imbalanced datasets with different dimensions or space complexity from the UCI Machine Learning Repository (Lichman, 2013) and other publicly available datasets (Maalouf & Siddiqi, 2014). More detailed information of the selected datasets is shown in Table 1. Additionally, to determine the mean and standard deviation of the performance measures for the oversampling methods, referencing (Nekooimehr & Lai-Yuen, 2016), 4-fold stratified cross validation was used. Each experiment was repeated 3 times to report the average to alleviate the randomness effects on the results.

4.2.1. In terms of parameter settings for the classifiers

We used four popular classifiers (Guo et al., 2017) of dealing with imbalanced classification to evaluate the oversampling methods: **1**) Support Vector Machine with radial basis function (RBF) (Suykens, Gestel, Brabanter, Moor, & Vandewalle, 2002) in Matlab 2012a, **2**) Artificial Neural Network (ANN), **3**) Random Forests (RF), and **4**) K-Nearest Neighbors (KNN) in (Yan, 2016). It should be noted that to ensure the generality of the results and highlight the differences of ten algorithms, therefore, the parameters of the four classifiers in this paper are not optimized. For the SVM classifiers, the radial basis parameters of the Wpbc, Ds1.100 and Ds 1.10 datasets are 5, and the radial basis parameters of the other 21 datasets are 0.5. For ANN, RF, and KNN three classifiers, we select the default settings for these classifiers in (Yan, 2016).

Table 1
Description of the imbalanced datasets.

No.	Dataset	Minority class	Majority class	Features	Instances	Minority instances	Majority instances	Imbalanced ratios
1	Wpbc	Class "N"	Class "R"	35	198	47	151	1/3.22
2	Vote	Class "republican"	Class "democrat"	16	435	168	267	1/1.59
3	Vehicle	Class "van"	All other	17	846	199	647	1/3.25
4	Segment	Class "WINDOW"	All other	18	2310 (1/5)	330 (1/5)	1980 (1/5)	1/6.01
5	Pima	Class "1"	Class "0"	8	768	268	500	1/1.87
6	Parkinsons	Class "0"	Class "1"	23	195	48	147	1/3.06
7	Iris	Class "2"	All other	4	150	50	100	1/2.00
8	Liver	Class "1"	All other	6	178	71	107	1/1.38
9	Lev	Class "1"	All other	4	1000	93	907	1/9.75
10	Leaf	Class "1-7"	All other	16	340	70	270	1/3.86
11	Heart	Class "1"	All other	13	270	120	150	1/1.25
12	Glass	Class "1"	All other	9	214	70	138	1/1.97
13	Haberman	Class "2"	Class "1"	3	306	81	225	1/2.78
14	Heating	Class "6-8"	All other	9	768	201	567	1/2.82
15	Breast	Class "car" and "fad"	All other	9	106	36	70	1/1.94
16	Balance	Class "B"	All other	4	625	288	337	1/1.17
17	yeast	Class "ME3, ME2, EXC,VAC, POX , ERL"	All other	8	1484	304	1180	1/3.89
18	New-thyroid1	Class "2"	All other	5	215	35	180	1/5.14
19	Page-blocks0	Class "text"	All other	10	5472 (1/4)	559 (1/4)	4913 (1/4)	1/8.79
20	Ecoli3	Class "imu"	All other	5	336	35	301	1/8.03
21	Abalone9-18	Class "18"	Class "9"	7	731	42	689	1/16.40
22	Musk	Class "-1"	All other	166	1319	203	1116	1/5.50
23	Ds1.100	Class "1"	Class "0"	100	267,33 (1/16)	804 (1/16)	2592,9 (1/16)	1/32.25
24	Ds1.10	Class "1"	Class "0"	10	267,33 (1/16)	804 (1/16)	2592,9 (1/16)	1/32.25

4.2.2. In terms of parameter settings for the sampling algorithms

Generally speaking, the optimal parameters of any oversampling algorithm cannot be obtained in advance for new unfamiliar datasets without prior experience. Thus, to highlight the insensitivity of the proposed NI-MWMOTE algorithm to the experience, this paper assumes that all algorithms can not obtain the optimal parameters of the 24 imbalanced datasets in advance. Therefore, according to the recommended range of the original literatures, we setted the parameters as follows:

For all methods, the k (number of nearest neighbors for SMOTE or knn-based noise processing method) values were setted between 3 and 5. For ADASYN and NI-MWMOTE, the k' (parameter of nearest neighbor density) values were setted to 5. For Cluster SMOTE and NI-MWMOTE, the m (number of minority sub-clusters) values were setted to 3 and 1–3, respectively. For A-SUWO and NI-MWMOTE, the K (fold number of cross-validation) values were setted to 3. For MWMOTE and NI-MWMOTE, the values of C_p , C_{fth} , C_{MAX} and k_2 (k_3) were all setted to 3, 5, 2 and 5, respectively. Additionally, the C_{thres} (coefficient to tune the threshold for clustering) values of A-SUWO were setted to 1, and the w values of NI-MWMOTE were setted to 0.5.

4.3. Comparison of the algorithms and results analysis

Tables 2–5 show the the mean results for the proposed NI-MWMOTE method and the other nine sampling methods on the 24 real imbalanced datasets classified using the four kinks of classifiers, where the best measures are highlighted in bold black.

4.3.1. Comparison of overall performances of algorithms

First, it is not difficult to find that when using SVM classifier, NI-MWMOTE obtains the best results at Sensitivity in 15 out of the 24 datasets, gains the best results at G-mean in 18 out of the 24 datasets, gets the best results at F-measure in 15 out of the 24 datasets, and acquires the best results at AUC in 19 out of the 24 datasets. The total winning rate is 69.79%. when using ANN classifier, NI-MWMOTE obtains the best results at Sensitivity in 7 out of the 24 datasets, gains the best results at G-mean in 9 out of the 24 datasets, gets the best results at F-measure in 8 out of the 24 data-

sets, and acquires the best results at AUC in 10 out of the 24 datasets. The total winning rate is 35.42%. when using RF classifier, NI-MWMOTE obtains the best results at Sensitivity in 17 out of the 24 datasets, gains the best results at G-mean in 16 out of the 24 datasets, gets the best results at F-measure in 12 out of the 24 datasets, and acquires the best results at AUC in 17 out of the 24 datasets. The total winning rate is 64.58%. when using KNN classifier, NI-MWMOTE obtains the best results at Sensitivity in 15 out of the 24 datasets, gains the best results at G-mean in 16 out of the 24 datasets, and acquires the best results at AUC in 17 out of the 24 datasets. The total winning rate is 60.42%.

Obviously, for all classifiers, the number of wins and the winning rate of NI-MWMOTE are the most in all algorithms, so it can be easily concluded that NI-MWMOTE has good performance on the most imbalanced datasets. Additionally, it is not difficult to see that the average values of NI-MWMOTE's all evaluation measures are optimal.

Next, the results of the imbalanced classification are further analyzed. Table 6 shows each algorithm's mean rankings in terms of Sensitivity, G-mean, F-measure and AUC for all the tested imbalanced datasets, where the best performing algorithm ranks 1 and the worst performing algorithm ranks 10. Obviously, according to Table 6, all the evaluation measures of NI-MWMOTE rank best. Additionally, Table 6 also shows the results of Friedman's test for each sampling algorithm. Friedman's test is a nonparametric equivalent of the repeated-measures ANOVA, and the null hypothesis in Friedman's test is whether all classifiers are performing similarly in mean rankings (Nekooimehr & Lai-Yuen, 2016). Therefore, analysis of Table 6 will not be difficult to find that, for all four classifiers and all four measures, there exists a significance level of $\alpha = 0.05$ to reject the null hypothesis, which means that classifiers do not perform similarly in mean rankings.

All the above null hypotheses are rejected for four performance measures, therefore, a post-hoc test is applied. Holm's test is the non-parametric analog of multiple t -test that adjusts α to compensate for multiple comparisons in a step-down procedure. Holm's null hypothesis is whether the proposed method performs better than others as the control algorithm (Douzas et al., 2018;

Table 2
Results (accuracy (%)) for the sampling methods on the 24 datasets classified using SVM.

(a) Sensitivity values (SEN)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	47.70	38.03	16.21	21.92	17.23	14.73	91.78	39.11	76.67	45.68	3.38	63.86	40.77	23.16	5.78	91.06	47.70	77.43	33.05	40.39	4.02	80.29	54.81	77.66	43.85
ROS	48.37	39.83	12.25	18.92	15.07	13.16	90.01	39.72	75.36	42.03	3.73	57.32	33.45	22.53	6.91	89.47	46.16	83.68	59.98	34.72	6.29	79.14	44.60	76.86	43.32
SMOTE	48.74	35.51	16.21	28.65	21.45	20.88	87.55	39.80	73.69	43.38	6.23	61.73	36.94	21.42	8.98	88.67	46.05	81.48	67.17	46.53	20.00	74.86	41.93	72.53	45.43
BSMOTE	50.79	37.77	6.14	5.36	13.64	12.74	92.04	40.24	79.05	41.66	6.00	54.88	39.22	19.88	9.35	90.67	40.13	61.81	46.97	34.26	12.80	76.17	32.64	70.57	40.62
ADASYN	46.55	38.96	16.05	15.09	20.33	17.86	89.30	42.79	79.05	46.03	5.40	62.44	42.65	22.46	5.31	89.66	46.60	63.66	70.90	43.29	17.50	78.49	35.42	73.88	44.57
Safe-level-SMOTE	43.69	37.38	14.96	24.02	19.62	18.75	88.12	37.28	73.33	37.64	6.09	58.73	31.10	19.07	7.34	90.24	44.41	82.18	61.96	41.55	3.18	73.73	31.73	63.85	42.08
Cluster-SMOTE	45.55	36.47	27.82	30.60	65.28	24.40	86.34	42.48	69.29	48.38	7.16	64.87	42.98	6.04	10.20	91.41	55.48	82.99	89.20	51.74	31.44	69.96	38.19	66.35	49.36
A-SUWO	33.28	38.43	9.63	7.97	10.49	12.00	89.66	36.99	74.90	31.48	3.96	58.85	26.98	6.04	2.38	89.52	39.25	70.14	23.04	27.78	0.76	71.07	25.32	51.67	35.07
MWMOTE	36.17	38.55	39.79	39.67	61.39	24.48	90.33	37.61	76.19	34.03	25.97	62.13	28.85	23.02	21.80	90.51	56.80	83.10	89.66	43.87	11.89	69.29	27.56	53.69	48.60
NI-MWMOTE	52.63	39.50	40.69	44.73	58.94	29.48	92.40	39.27	76.43	45.24	29.61	68.04	39.32	26.04	47.02	91.58	60.75	93.17	93.28	55.21	42.42	73.02	39.96	72.01	56.28
(b) G-mean values (G-M)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	62.45	61.75	40.53	47.11	39.71	39.27	95.10	57.60	76.26	66.49	18.47	76.89	53.44	35.96	23.28	92.97	67.39	87.17	56.37	60.44	12.76	88.85	73.20	80.07	58.90
ROS	61.89	62.69	35.05	43.57	37.36	36.79	94.85	58.54	75.88	63.90	19.41	72.46	49.69	37.68	25.57	92.75	66.30	90.62	76.34	54.03	15.84	88.34	66.14	80.66	58.60
SMOTE	64.38	59.77	40.33	54.00	43.80	46.00	93.32	58.09	75.61	64.78	24.88	75.29	52.29	37.31	28.74	92.44	66.15	89.28	80.27	65.92	43.02	86.01	64.19	79.13	61.88
BSMOTE	64.61	61.56	24.82	23.53	35.48	36.38	95.63	58.09	73.29	63.55	24.39	70.42	53.08	34.75	29.14	92.57	61.20	77.72	67.15	53.11	30.90	86.60	54.05	77.65	56.24
ADASYN	63.09	62.52	40.10	39.65	42.46	43.17	94.65	59.70	76.18	67.25	23.28	75.13	55.35	37.29	22.20	92.65	65.75	78.75	82.20	62.15	38.38	87.94	58.60	78.96	60.31
Safe-level-SMOTE	62.39	61.30	38.80	49.55	42.44	44.00	94.06	56.76	75.40	61.04	24.74	73.98	48.70	36.67	26.16	92.57	65.43	89.96	77.51	59.32	8.73	85.36	54.65	76.19	58.57
Cluster-SMOTE	61.64	60.46	52.80	55.54	70.52	49.72	93.29	59.28	74.57	68.73	26.49	76.60	55.28	22.62	30.46	92.76	71.43	90.33	89.99	69.27	51.43	83.38	61.00	75.79	64.31
A-SUWO	54.99	62.13	31.12	28.54	31.56	34.97	94.49	56.32	75.30	56.02	19.97	74.12	46.75	22.62	15.16	92.20	61.65	82.44	47.30	49.44	2.50	83.86	50.00	69.47	51.79
MWMOTE	57.01	62.19	63.17	63.59	70.39	50.35	94.79	57.19	76.11	58.60	50.42	75.18	47.66	45.34	43.88	92.84	72.63	90.28	89.74	63.86	27.34	82.92	51.37	70.66	64.90
NI-MWMOTE	67.10	62.97	63.71	67.57	70.01	54.90	95.73	58.59	76.97	66.89	53.58	77.80	54.53	46.13	59.92	93.49	73.96	96.09	90.56	71.49	62.69	85.27	61.63	80.06	70.49
(c) F-measure values (F-M)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	45.71	55.19	28.42	36.78	26.02	NaN	93.60	48.95	64.29	57.14	NaN	70.30	36.80	18.32	NaN	92.43	57.66	84.06	45.33	43.88	NaN	84.87	59.53	21.71	44.62
ROS	44.75	55.10	22.12	32.04	23.63	NaN	93.63	50.14	64.02	54.26	NaN	65.39	32.64	19.54	NaN	92.25	56.76	88.09	65.89	NaN	NaN	84.97	54.67	23.60	42.65
SMOTE	49.00	52.61	28.17	45.31	30.50	35.42	91.70	49.64	63.82	55.36	11.95	68.86	35.68	19.10	NaN	91.91	56.04	86.85	68.37	50.88	22.25	82.90	53.86	25.39	48.98
BSMOTE	48.66	54.95	12.13	NaN	21.51	NaN	94.27	49.76	60.60	53.54	11.48	62.40	36.36	16.96	14.72	91.99	48.75	73.07	54.72	NaN	NaN	82.66	NaN	22.80	37.97
ADASYN	46.88	56.18	27.80	NaN	28.86	NaN	93.21	51.73	63.99	58.04	NaN	68.26	39.02	19.19	NaN	92.12	53.69	74.29	68.88	46.46	NaN	84.32	47.27	23.08	43.47
Safe-level-SMOTE	47.33	54.58	26.62	40.00	29.26	NaN	92.64	47.97	63.54	52.23	NaN	67.49	31.91	18.95	NaN	92.03	56.33	87.70	67.69	NaN	NaN	82.13	44.78	30.29	43.06
Cluster-SMOTE	45.63	53.52	43.57	47.19	62.75	39.93	91.73	51.32	63.08	59.70	13.41	69.94	38.65	8.45	15.85	92.21	59.77	88.10	66.13	54.40	NaN	80.87	49.49	23.31	50.79
A-SUWO	41.08	55.66	18.20	NaN	17.71	NaN	93.43	47.31	64.30	46.20	NaN	67.52	30.15	8.45	NaN	91.63	52.21	78.74	34.89	NaN	NaN	80.77	39.47	33.78	37.56
MWMOTE	41.93	55.76	56.41	56.83	63.84	NaN	93.38	48.44	64.20	48.86	59.10	68.49	31.24	29.67	29.55	92.29	61.77	87.41	64.46	49.04	NaN	80.41	41.07	32.30	52.35
NI-MWMOTE	53.16	56.75	57.07	61.38	63.96	46.02	94.70	50.33	65.33	57.90	68.83	70.81	38.19	28.98	48.28	93.01	61.04	94.50	62.57	56.15	35.64	83.02	50.21	28.80	59.44
(d) AUC values (AUC)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	64.73	69.15	58.77	61.58	54.37	59.72	95.16	61.97	76.30	71.23	52.16	78.22	55.41	39.50	49.79	92.99	71.60	88.16	65.93	68.09	50.92	89.34	76.97	80.34	68.02
ROS	63.78	69.25	56.27	59.62	53.84	57.99	94.98	63.00	75.99	69.59	52.39	74.46	53.63	42.77	50.76	92.81	71.08	91.37	78.76	64.92	52.42	88.93	72.03	80.92	67.98
SMOTE	66.89	68.06	58.28	65.21	55.45	61.11	93.51	62.29	75.69	70.06	52.83	76.78	55.48	43.20	50.48	92.52	70.71	90.37	82.01	71.27	58.21	86.94	70.77	79.95	69.09
BSMOTE	66.49	69.05	53.27	54.30	52.97	58.33	95.70	62.05	73.60	69.30	52.61	72.62	55.53	40.31	50.07	92.59	66.99	80.34	72.17	65.41	54.88	87.35	66.08	78.16	66.26
ADASYN	66.03	69.64	58.13	59.65	54.51	61.11	94.81	63.04	76.33	72.14	52.92	76.42	57.24	42.19	49.04	92.70	69.88	81.27	83.53	69.43	56.62	88.56	67.52	79.59	68.43
Safe-level-SMOTE	66.39	68.95	57.80	63.11	55.71	61.00	94.26	61.85	75.51	68.31	53.28	75.96	53.68	44.79	50.29	92.60	70.54	90.72	79.83	69.12	51.37	86.36	65.79	77.84	68.13
Cluster-SMOTE	64.48	68.35	64.02	65.70	70.73	62.85	93.57	62.60	74.85	73.01	52.60	77.66	57.04	45.37	50.59	92.77	73.93	91.12	90.02	73.99	63.40	84.71	68.87	76.83	70.79
A-SUWO	62.07	69.44	55.11	55.10	52.71	56.94	94.62	61.37	75.60	65.59	52.28	76.10	53.99	45.37	49.										

Table 3
Results (accuracy (%)) for the sampling methods on the 24 datasets classified using ANN.

(a) Sensitivity values (SEN)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9–18	Musk	S1.100	S1.10	Average
None Sampling	31.08	93.28	72.30	71.01	30.44	74.54	91.54	3.90	53.05	0.99	73.17	94.71	0.14	1.12	33.36	92.50	4.61	43.17	37.37	27.66	2.42	81.32	56.46	0.64	44.62
ROS	51.85	95.06	79.49	86.83	63.25	81.41	79.60	37.19	71.92	37.59	76.11	99.08	41.16	27.80	45.10	91.80	79.39	82.06	69.07	94.33	68.33	85.38	74.20	90.81	71.20
SMOTE	50.24	94.67	78.51	83.64	66.32	81.83	79.85	38.94	72.68	41.35	73.48	97.58	41.59	28.39	47.27	91.65	82.24	88.43	70.29	94.21	74.85	87.02	84.51	91.56	72.55
BSMOTE	50.05	95.48	77.93	75.12	69.49	81.49	78.01	38.69	61.38	35.18	73.24	99.66	50.33	24.91	48.46	91.96	88.16	97.22	81.02	94.10	74.17	89.49	82.69	94.87	73.05
ADASYN	50.68	94.13	76.26	77.95	67.90	77.72	77.55	29.11	66.97	36.76	73.18	99.89	45.98	23.33	49.98	91.95	82.57	98.03	79.15	94.33	71.29	87.37	84.13	93.48	72.07
Safe-level-SMOTE	52.23	94.41	78.00	86.64	64.17	78.43	78.49	36.62	72.35	35.75	69.86	99.55	48.67	24.61	46.19	92.19	81.58	83.68	71.50	94.33	65.61	87.19	74.47	80.98	70.73
Cluster-SMOTE	48.67	95.56	76.63	87.02	69.92	79.07	78.41	29.08	70.91	39.86	73.29	98.99	46.19	28.34	40.84	90.64	83.66	89.47	76.48	94.10	68.26	84.25	85.26	94.02	72.04
A-SUWO	50.87	95.83	78.81	77.92	67.18	77.98	77.48	42.82	71.51	20.50	74.58	96.54	39.36	25.32	50.86	90.69	68.09	86.46	75.71	94.21	62.12	82.42	73.56	79.81	69.19
MWMOTE	50.06	95.14	75.52	83.68	70.35	71.33	77.50	36.27	74.11	28.18	75.66	100.00	36.64	25.51	40.27	90.82	84.98	93.17	76.74	94.21	60.23	81.79	70.57	82.10	69.78
NI-MWMOTE	MWMOTE55.7795.8978.7780.5367.9381.0679.5942.6371.4342.0276.2399.9542.7323.1143.3791.9886.5197.1179.1294.4477.0585.8786.0094.0773.88																								
(b) G-mean values (G-M)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9–18	Musk	S1.100	S1.10	Average
None Sampling	52.90	91.35	84.26	84.91	53.97	86.44	94.85	19.59	68.22	9.84	78.84	48.41	3.73	10.62	54.41	95.22	20.94	64.87	60.85	48.59	7.78	87.49	74.43	2.31	54.37
ROS	60.00	91.98	88.93	91.58	71.52	88.74	89.69	54.44	76.86	45.91	80.01	63.25	56.42	43.12	57.74	94.51	84.19	90.35	81.64	87.05	71.34	89.13	81.64	78.23	75.76
SMOTE	60.11	91.92	88.38	90.61	71.96	89.07	89.72	54.83	77.35	46.25	78.33	62.14	55.92	40.42	58.48	94.00	84.81	93.68	82.39	87.35	70.74	89.76	81.91	76.15	75.68
BSMOTE	60.45	92.10	87.84	86.55	73.57	90.33	88.61	54.52	73.11	41.94	79.29	63.16	57.21	37.96	56.93	94.67	81.57	97.54	86.56	84.85	68.81	90.67	78.78	70.42	74.89
ADASYN	59.04	91.06	87.44	88.40	72.56	88.59	88.26	49.93	75.40	44.14	78.85	62.86	57.80	39.62	59.93	94.30	83.48	97.77	86.25	86.11	66.57	89.71	79.13	71.12	74.93
Safe-level-SMOTE	61.31	91.66	88.07	91.20	71.24	87.11	88.97	53.15	76.69	42.97	77.70	62.48	58.68	39.54	58.20	94.11	84.72	91.26	83.04	87.57	67.75	89.81	81.23	76.48	75.21
Cluster-SMOTE	59.30	92.17	87.21	90.75	73.47	86.62	89.54	47.84	76.10	46.78	76.71	62.72	56.23	47.85	58.57	93.94	84.36	94.39	85.61	86.49	68.59	88.82	81.09	68.38	75.15
A-SUWO	61.25	92.17	88.20	87.43	71.96	87.16	88.40	56.79	75.65	38.06	79.31	71.17	52.76	40.43	61.58	93.70	76.70	92.33	82.82	87.46	61.67	88.11	79.71	76.33	74.63
MWMOTE	60.36	91.24	86.58	89.47	73.49	82.60	88.84	53.32	77.33	40.47	79.74	63.17	52.99	45.92	57.86	93.52	84.45	96.08	85.75	87.76	62.32	87.57	77.43	74.19	74.69
NI-MWMOTE	63.01	92.23	88.33	89.14	71.97	89.11	89.84	57.23	76.21	48.37	80.33	65.75	58.20	45.04	60.59	94.69	84.49	98.04	86.96	87.42	67.94	89.73	82.22	70.56	76.56
(c) F-measure values (F-M)																									
Method	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9–18	Musk	S1.100	S1.10	Average
None Sampling	38.24	89.03	80.59	82.76	43.93	84.56	92.19	NaN	57.47	NaN	77.13	36.87	NaN	NaN	41.48	94.91	8.61	59.06	53.27	NaN	NaN	76.55	60.85	NaN	44.90
ROS	41.69	89.63	74.90	71.96	63.59	77.12	84.14	58.21	64.78	30.40	78.50	61.86	40.89	24.72	43.73	94.11	71.97	89.69	70.34	52.72	24.02	76.60	31.24	14.91	59.66
SMOTE	42.79	89.54	74.06	69.40	64.05	77.84	84.11	58.67	65.38	31.90	76.92	60.97	39.96	22.57	44.96	93.54	71.35	92.90	70.60	53.06	21.20	76.53	20.26	13.61	59.01
BSMOTE	43.46	89.77	73.28	57.50	66.19	78.82	82.61	57.91	60.87	29.22	78.23	62.11	41.20	20.04	43.58	94.28	62.51	93.78	66.09	48.19	19.78	76.59	17.00	11.16	57.26
ADASYN	41.60	88.50	72.56	60.76	64.78	76.04	82.11	40.59	63.05	29.42	71.35	62.06	43.08	21.35	46.89	93.86	68.16	93.39	68.88	50.25	18.73	75.95	16.88	11.38	56.73
Safe-level-SMOTE	43.53	89.27	73.47	72.31	63.18	75.24	83.15	57.71	64.58	28.07	76.34	61.70	42.61	21.44	44.17	93.65	71.61	90.44	71.40	53.89	21.15	76.52	29.22	15.63	59.18
Cluster-SMOTE	42.09	89.84	72.21	71.22	65.87	75.07	84.59	56.49	63.82	32.61	74.96	61.78	40.37	30.57	46.04	93.51	69.34	93.78	71.88	51.57	22.55	76.99	18.88	10.62	59.03
A-SUWO	43.60	89.81	73.97	62.06	64.10	74.43	82.49	50.59	63.38	20.66	77.70	65.95	36.23	22.46	48.24	93.22	62.65	90.68	60.79	53.54	16.39	77.02	28.14	15.94	57.25
MWMOTE	42.22	88.66	71.28	68.73	65.97	68.46	83.49	58.06	65.48	25.79	78.00	62.29	36.54	28.96	44.65	93.03	68.73	94.45	72.21	54.35	18.00	75.99	23.45	13.78	58.44
NI-MWMOTE	45.92	89.98	73.98	65.54	64.11	78.85	84.60	59.23	64.09	33.52	78.96	63.58	43.04	29.03	48.16	94.26	68.19	96.22	73.11	53.20	18.58	78.16	19.98	11.16	59.81
(d) AUC values (AUC)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9–18	Musk	S1.100	S1.10	Average
None Sampling	60.56	91.37	85.25	86.27	63.06	87.39	94.91	51.20	70.39	49.38	79.06	60.09	49.82	51.04	61.05	95.26	52.09	71.59	68.56	63.22	51.21	87.84	77.78	50.32	69.11
ROS	60.64	92.03	89.49	91.71	72.06	89.07	90.33	58.44	77.03	46.83	80.11	70.02	59.25	47.34	59.51	94.55	84.37	91.03	83.03	87.36	71.95	89.26	82.26	79.21	76.95
SMOTE	61.08	91.96	89.00	90.90	72.20	89.39	90.33	58.07	77.50	46.54	78.49	68.87	58.39	42.97	59.81	94.03	84.86	94.12	83.52	87.69	71.14	89.84	82.09	77.54	76.68
BSMOTE	61.53	92.16	88.47	87.42	73.69	90.81	89.33	57.76	74.23	42.59	79.54	70.14	57.68	41.38	57.67	94.71	81.92	97.59	86.85	85.37	69.18	90.71	78.96	73.63	75.97
ADASYN	59.73	91.11	88.26	89.10	72.72	89.35	89.00	57.38	75.93	44.88	79.07	70.02	59.32	45.31	60.92	94.33	83.56	97.81	86.70	86.53	67.13	89.79	79.57	73.87	76.31
Safe-level-SMOTE	62.10	91.70	88.72	91.32	71.63	87.59	89.67	56.88	76.82	43.70	78.14	69.68	59.71	44.07	59.76	94.13	84.81	91.75	84.15	87.92	68.36	89.87	81.94	76.91	76.31
Cluster-SMOTE	60.46	92.23	87.94	90.83	73.56	86.98	90.33	53.89	76.29	47.38	76.79	69.66	57.32	54.57	62.42	94.00	84.42	94.64	86.18	86.86	69.13	88.98	81.36	72.04	76.59
A-SUWO	62.31	92.24	88.76	88.01	72.13	87.70	89.17	59.07	75.77	45.58	79.46	74.73	55.04	44.94	62.71	93.75	78.10	92.95	83.58	87.757					

Table 4
Results (accuracy (%)) for the sampling methods on the 24 datasets classified using RF.

(a) Sensitivity values (SEN)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	17.07	91.99	92.11	85.80	57.89	66.42	90.77	57.41	47.74	0.12	74.15	78.02	25.54	25.14	33.61	86.10	69.52	92.59	69.51	45.95	10.08	61.91	25.81	23.50	55.36
ROS	30.16	94.06	94.01	86.17	64.56	72.40	88.88	59.63	60.55	8.58	76.01	78.42	34.13	26.33	38.97	86.75	73.25	90.63	75.31	55.21	13.64	71.59	27.88	34.62	60.07
SMOTE	37.30	94.45	94.00	89.81	69.41	78.60	89.94	60.73	65.06	17.76	75.57	78.06	42.34	29.94	41.13	87.93	78.18	91.32	83.48	61.57	41.44	77.49	35.04	40.44	65.04
BSMOTE	35.81	93.32	92.76	89.20	67.73	81.22	92.28	59.07	61.70	17.12	74.50	74.02	44.05	28.54	42.36	86.32	76.10	92.25	78.92	58.45	35.68	71.27	23.29	41.13	63.21
ADASYN	34.81	92.81	94.50	89.30	69.91	83.20	90.91	61.20	66.11	17.19	75.02	76.11	47.19	29.45	41.11	86.74	79.28	93.40	86.06	66.90	42.88	76.95	31.36	43.75	65.67
Safe-level-SMOTE	26.40	93.30	93.55	87.44	67.11	71.77	91.21	62.30	61.38	12.72	73.36	75.09	40.33	26.14	39.75	86.16	76.54	91.44	79.63	62.62	18.41	77.99	35.26	28.21	61.59
Cluster-SMOTE	36.69	93.48	92.81	87.61	67.99	69.80	92.69	61.63	60.43	20.24	74.08	80.49	48.29	26.62	39.60	87.53	80.15	93.29	86.32	67.01	51.67	68.62	30.50	36.54	64.75
A-SUWO	20.76	93.48	93.11	85.29	62.54	80.68	87.02	58.59	55.53	4.67	74.05	78.31	34.36	25.20	33.68	87.29	72.15	93.29	70.22	51.62	6.44	68.05	29.22	28.69	58.09
MWMOTE	33.40	93.10	93.33	88.43	67.94	75.66	89.78	55.97	60.56	15.92	74.07	74.87	33.30	24.65	34.69	87.26	78.18	91.32	88.52	56.37	24.62	74.56	37.23	34.78	62.02
NI-MWMOTE	39.35	93.79	94.85	90.91	70.26	83.26	93.13	62.25	62.02	25.04	76.44	79.93	45.60	26.80	44.77	88.90	81.69	93.40	91.34	75.12	60.15	74.55	38.41	49.09	68.38
(b) G-mean values (G-M)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	41.82	95.31	95.15	92.85	70.62	80.77	94.33	68.60	64.27	3.32	81.25	84.86	48.03	49.83	55.75	89.57	81.22	95.94	82.59	65.68	25.36	78.14	49.80	44.85	68.33
ROS	54.18	95.43	95.38	93.48	72.67	83.88	94.21	68.96	70.45	26.92	81.66	84.02	52.58	49.99	57.80	89.98	82.69	94.87	85.77	72.05	32.84	84.06	51.59	57.60	72.21
SMOTE	58.95	95.82	95.94	95.17	74.27	87.10	94.96	68.60	72.80	36.74	80.79	83.79	56.65	51.87	59.76	90.58	84.67	95.18	89.68	75.21	60.70	87.16	58.06	61.99	75.69
BSMOTE	57.66	95.25	95.12	94.37	72.24	89.58	95.32	68.79	69.94	35.87	80.23	81.22	57.52	50.35	60.42	89.26	82.50	95.39	87.09	72.82	55.06	83.63	46.58	62.02	74.09
ADASYN	57.00	95.53	95.87	94.30	73.65	88.67	94.96	69.91	72.72	35.37	81.50	84.20	59.97	51.53	59.56	89.97	84.29	96.07	90.49	78.03	62.80	86.63	54.40	64.43	75.91
Safe-level-SMOTE	50.58	95.23	95.48	94.18	73.77	83.99	94.78	70.27	70.52	31.28	80.42	82.94	57.18	49.74	58.58	89.38	83.90	95.31	87.90	75.99	41.38	87.15	58.60	51.64	73.34
Cluster-SMOTE	56.82	95.41	95.17	93.96	73.94	81.73	95.73	69.56	71.42	37.73	80.75	84.89	59.37	50.69	58.89	89.82	84.66	96.18	90.28	78.03	68.30	82.39	51.25	57.99	75.21
A-SUWO	46.01	94.85	95.47	92.79	71.83	87.40	93.35	68.57	68.74	20.57	80.49	84.04	53.15	49.71	55.56	89.58	82.00	96.09	82.77	68.70	17.65	82.04	53.27	52.55	70.30
MWMOTE	56.33	95.03	95.26	94.52	74.32	84.27	94.42	67.07	71.45	34.78	80.62	82.71	53.49	49.40	56.18	89.55	83.86	94.98	91.50	72.05	47.70	85.80	60.57	57.70	73.90
NI-MWMOTE	60.97	95.73	96.23	95.38	74.53	88.88	95.49	70.37	71.14	41.41	81.69	85.72	59.96	50.83	61.65	90.79	84.53	96.21	92.08	83.37	73.38	85.70	61.25	67.77	77.71
(c) F-measure values (F-M)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	NaN	94.09	92.49	92.35	62.90	76.57	92.11	63.18	52.40	NaN	79.64	80.09	33.91	38.89	44.31	88.81	73.63	95.03	75.57	56.12	NaN	74.64	39.93	NaN	58.61
ROS	42.84	94.17	91.38	92.55	65.03	77.56	92.27	63.83	58.54	NaN	79.92	78.35	36.80	37.08	44.99	89.24	73.67	93.92	77.50	56.62	NaN	80.56	42.02	42.98	62.99
SMOTE	45.70	94.65	91.73	93.22	66.92	80.91	93.27	63.77	61.15	19.93	79.01	77.92	40.55	37.25	47.44	89.91	74.46	93.95	77.82	56.52	36.69	72.52	45.90	37.35	65.77
BSMOTE	44.42	93.83	90.83	92.23	64.56	83.87	93.71	63.75	57.39	18.78	78.61	74.71	41.28	35.18	48.32	89.31	70.20	93.12	74.35	56.34	NaN	78.80	33.64	38.78	63.17
ADASYN	43.71	94.12	91.41	89.59	66.11	81.72	92.87	65.10	60.63	18.35	79.73	78.57	44.48	37.16	47.03	89.20	72.32	94.18	75.69	58.48	34.30	81.01	42.77	37.22	65.66
Safe-level-SMOTE	38.32	93.94	91.39	92.94	66.31	77.01	93.09	65.37	58.29	NaN	78.71	76.86	41.80	36.62	45.93	88.57	73.88	94.09	77.66	58.68	27.10	81.30	50.03	38.19	64.42
Cluster-SMOTE	42.18	94.23	90.53	92.26	66.44	72.15	94.23	64.90	60.13	20.24	79.08	78.69	43.38	38.73	46.49	89.02	72.55	94.65	73.76	58.08	37.70	79.24	NaN	32.52	63.38
A-SUWO	NaN	93.47	91.06	91.07	64.09	81.23	91.38	63.28	57.03	NaN	78.70	78.21	37.51	38.40	43.98	88.78	72.94	94.25	74.32	52.10	NaN	78.44	43.67	40.67	60.61
MWMOTE	43.94	93.75	90.58	92.12	67.09	74.28	92.29	61.36	60.20	18.06	78.86	76.67	38.87	38.55	43.98	88.71	72.07	92.76	75.32	54.79	27.58	82.41	50.92	40.25	64.81
NI-MWMOTE	48.23	94.63	91.83	93.08	67.16	83.04	93.87	66.00	59.21	23.90	80.04	80.68	44.76	38.85	49.39	90.10	71.01	94.65	72.03	64.92	39.56	81.83	48.84	35.89	67.23
(d) AUC values (AUC)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None Sampling	59.77	95.37	95.20	93.14	72.02	82.32	94.40	69.69	67.13	47.23	81.59	85.16	57.93	61.95	63.04	89.64	82.27	96.11	83.95	72.20	55.01	80.71	62.93	61.75	75.44
ROS	63.74	95.44	95.39	93.79	73.18	84.79	94.37	69.69	71.26	46.54	81.87	84.22	57.57	60.62	62.35	90.04	83.33	95.13	86.57	75.33	56.31	85.27	63.91	66.93	76.57
SMOTE	65.23	95.83	95.96	95.33	74.44	87.56	95.10	69.11	73.26	46.88	80.97	84.00	59.07	59.90	63.98	90.62	84.97	95.38	89.98	77.52	68.25	87.81	67.31	68.98	78.23
BSMOTE	64.33	95.27	95.15	94.52	72.39	90.01	95.37	69.59	70.49	46.13	80.45	81.57	59.58	58.68	64.27	89.31	82.79	95.57	87.58	76.84	65.42	84.77	61.49	69.54	77.13
ADASYN	64.07	95.57	95.88	94.44	73.75	88.85	95.05	70.53	73.05	44.99	81.78	84.63	61.70	59.81	63.70	90.03	84.48	96.23	90.68	80.07	68.25	87.30	65.48	70.46	78.37
Safe-level-SMOTE	61.65	95.25	95.50	94.44	74.10	85.03	94.85	70.78	71.20	44.82	80.76	83.35	60.70	60.39	63.04	89.44	84.31	95.44	88.38	78.48	58.79	87.72	67.58	63.86	77.08
Cluster-SMOTE	62.34	95.43	95.20	94.19	74.20	82.75	95.78	70.07	72.42	45.29	81.05	85.01	60.64	61.57	63.59	89.85	84.82	96.27	90.44	79.80	72.25	83.94	64.91	66.97	77.87
A-SUWO	61.37	94.86	95.50	93.12	72.52	87.68	93.58																		

Table 5
Results (accuracy (%)) for the sampling methods on the 24 datasets classified using KNN.

(a) Sensitivity values (SEN)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None	33.78	88.52	84.78	84.26	53.76	89.59	88.51	49.30	51.50	27.00	71.00	69.56	26.39	20.12	46.81	85.01	63.49	97.22	66.70	53.13	14.85	82.61	59.19	47.76	60.62
Sampling																									
ROS	38.04	88.10	85.51	83.83	54.90	88.91	89.79	52.65	52.95	21.24	71.84	71.56	29.68	19.54	44.69	83.87	63.93	95.14	64.29	52.43	14.47	82.58	60.04	46.31	60.68
SMOTE	49.13	88.58	88.30	88.88	57.47	87.44	90.03	51.25	55.59	36.45	72.29	71.51	31.99	21.86	50.86	84.01	67.11	99.07	77.46	60.76	41.06	87.04	70.19	55.98	66.01
BSMOTE	47.09	89.14	89.37	88.84	56.71	88.83	89.70	55.31	54.02	34.47	71.76	74.56	30.00	21.23	48.49	85.78	66.12	96.18	75.06	59.26	34.09	86.04	65.12	54.81	65.08
ADASYN	43.75	88.03	89.23	90.14	59.05	92.69	85.17	56.01	56.08	35.30	70.93	73.18	37.18	22.34	47.32	84.56	69.19	99.07	77.90	67.94	38.71	86.20	70.14	50.48	66.27
Safe-level-SMOTE	42.51	88.07	88.18	87.19	56.32	88.35	91.80	54.82	54.50	34.40	71.14	72.19	32.87	21.06	46.44	85.53	66.56	97.22	72.70	61.81	19.85	85.70	64.64	53.63	64.06
Cluster-SMOTE	51.47	87.70	89.83	90.58	61.07	87.60	90.89	53.42	51.81	35.95	72.62	70.75	44.26	22.86	48.46	84.74	74.12	97.11	81.77	73.03	39.70	83.91	72.01	56.20	67.58
A-SUWO	43.58	88.22	86.62	86.73	56.50	91.34	88.39	50.93	55.07	21.34	72.67	68.85	28.49	19.89	39.37	83.20	64.25	90.51	69.29	61.46	11.97	80.60	58.60	39.96	60.74
MWMOTE	56.17	88.66	91.04	89.74	62.31	86.48	91.11	49.15	54.71	24.33	71.63	69.97	29.15	21.97	36.40	85.64	74.01	97.11	83.95	77.31	23.71	81.94	59.24	48.50	64.76
NI-MWMOTE	54.34	89.10	91.72	91.07	61.79	92.08	91.85	56.85	51.85	39.63	71.71	75.50	34.86	23.56	48.40	86.62	77.63	100.00	85.13	78.13	59.32	86.72	73.45	62.77	70.17
(b) G-mean values (G-M)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None	51.82	90.82	90.15	91.49	65.17	93.38	94.15	58.72	64.50	45.06	77.08	78.77	44.99	38.92	61.32	87.62	75.95	97.92	80.66	70.30	32.48	88.25	75.71	67.91	71.80
Sampling																									
ROS	54.61	91.35	90.58	91.15	65.99	93.90	94.98	61.36	65.80	40.56	75.99	78.22	48.54	38.52	60.28	87.30	76.24	96.90	79.20	69.25	31.58	88.18	76.51	67.03	71.83
SMOTE	58.74	91.56	92.29	93.87	66.88	92.39	94.68	59.29	66.50	50.16	76.46	78.58	48.71	41.16	62.49	87.19	77.04	98.77	86.29	73.99	60.76	89.82	80.45	72.61	75.03
BSMOTE	59.05	91.45	92.64	93.83	65.43	93.45	94.71	60.35	65.50	48.01	76.08	79.44	47.51	40.21	60.98	88.40	75.49	97.48	85.14	73.52	54.89	89.50	77.69	72.18	74.29
ADASYN	55.36	91.28	91.90	94.39	67.25	95.73	93.73	61.61	66.35	48.82	75.39	78.85	52.39	41.31	60.30	87.47	77.34	98.78	86.19	79.22	58.53	89.50	79.90	68.62	75.01
Safe-level-SMOTE	56.91	91.58	91.69	92.63	66.39	92.34	94.83	60.56	65.89	49.10	75.44	79.79	50.01	40.05	59.83	87.38	77.66	98.01	84.03	75.31	41.66	89.19	78.81	71.80	73.79
Cluster-SMOTE	59.20	90.66	92.83	94.62	67.51	91.57	95.36	58.89	64.52	47.42	76.43	78.73	55.15	41.60	62.17	87.34	79.12	97.64	87.57	81.67	58.93	88.56	81.10	72.00	75.44
A-SUWO	59.80	91.47	90.78	92.91	66.92	94.65	93.71	59.31	67.62	40.49	78.58	76.46	47.45	41.96	57.16	85.78	77.14	94.15	82.21	75.64	23.47	87.18	74.49	61.80	71.71
MWMOTE	64.10	91.92	92.95	94.38	69.71	92.06	94.93	58.91	67.61	41.04	77.61	76.97	48.99	44.61	55.07	89.10	80.64	97.74	88.40	84.74	46.68	88.08	74.16	68.20	74.53
NI-MWMOTE	61.64	92.08	94.07	95.12	68.08	96.37	95.68	62.07	65.09	50.89	77.58	80.57	53.02	42.26	62.24	89.25	81.19	99.34	88.51	84.24	72.30	89.89	81.36	75.41	77.43
(c) F-measure values (F-M)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None	34.15	88.51	84.99	86.85	55.76	88.66	92.62	52.30	51.25	28.00	75.32	71.66	28.03	20.89	48.24	86.65	63.96	95.41	72.30	53.33	NaN	77.35	52.79	46.81	60.66
Sampling																									
ROS	36.78	89.05	85.56	85.49	56.77	89.32	93.60	55.24	52.97	24.08	73.96	70.64	31.89	20.51	47.18	86.35	64.41	94.71	69.99	50.41	NaN	77.13	55.23	46.33	60.73
SMOTE	40.27	89.29	86.87	87.11	57.87	87.76	92.92	53.19	53.39	32.98	74.82	71.02	31.72	23.20	49.17	86.26	63.50	95.85	73.93	52.75	30.79	76.78	35.00	38.96	61.89
BSMOTE	40.88	89.20	87.51	88.27	56.07	88.63	92.88	54.94	52.23	30.83	74.52	71.91	30.59	22.21	47.63	87.58	60.41	95.34	73.60	53.63	25.30	76.93	36.01	39.64	61.53
ADASYN	36.82	88.87	85.78	86.94	58.28	90.76	91.86	56.42	53.09	31.69	73.45	71.35	35.62	23.32	46.66	86.54	62.49	95.90	71.55	60.21	28.07	76.72	34.14	32.76	61.64
Safe-level-SMOTE	39.47	89.46	85.86	86.04	57.28	86.47	93.09	55.16	52.75	31.95	73.82	72.73	31.91	21.99	46.14	86.43	65.72	95.80	74.14	56.13	NaN	76.33	47.86	47.36	61.41
Cluster-SMOTE	40.81	88.22	86.46	88.79	58.67	83.67	93.63	53.80	51.20	30.38	74.60	71.03	38.60	23.55	49.04	86.57	63.31	94.51	69.65	60.70	30.31	76.56	33.24	30.46	61.57
A-SUWO	43.09	89.17	85.07	87.92	57.92	90.64	91.81	53.20	55.33	23.13	76.53	68.67	30.68	25.86	43.90	84.79	66.73	90.80	73.54	60.73	NaN	76.20	47.57	45.26	61.19
MWMOTE	46.52	89.67	85.82	88.97	61.37	84.34	93.16	52.18	55.33	23.66	75.85	69.27	32.88	29.35	41.53	88.48	67.10	94.85	68.96	67.16	31.20	77.72	36.98	41.29	62.65
NI-MWMOTE	43.54	89.89	86.86	89.58	59.45	91.53	94.11	57.08	52.08	34.01	75.85	73.51	37.38	24.10	49.15	88.57	65.87	96.88	66.88	62.26	36.81	77.51	31.24	28.50	63.03
(d) AUC values (AUC)																									
Methods	Wpbc	Vote	Vehicle	Segment	Pima	Parkinsons	Iris	Liver	Lev	Leaf	Heart	Glass	Haberman	Heating	Breast	Balance	Yeast	New-thyroid1	Page-blocks0	Ecoli3	Abalone 9-18	Musk	S1.100	S1.10	Average
None	56.64	90.85	90.32	91.80	66.38	93.46	94.33	59.62	66.14	51.10	77.34	79.38	51.54	47.71	63.57	87.66	77.22	97.96	82.39	73.96	56.36	88.50	78.57	73.00	74.83
Sampling																									
ROS	58.22	91.41	90.73	91.47	67.11	94.04	95.13	62.08	67.36	49.35	76.11	78.53	54.53	47.74	63.00	87.37	77.51	97.01	81.06	73.17	56.00	88.41	79.13	72.31	74.95
SMOTE	59.68	91.61	92.38	94.01	67.65	92.53	94.80	59.92	67.57	52.74	76.58	78.93	53.08	49.68	63.82	87.25	77.86	98.79	86.91	76.51	66.71	89.89	81.52	75.96	76.52
BSMOTE	60.57	91.48	92.70	93.97	66.10	93.57	94.85	60.58	66.72	50.67	76.21	79.60	52.62	48.70	62.59	88.44	76.21	97.53	85.87	76.08	63.18	89.60	79.50	75.46	75.95
ADASYN	56.90	91.34	91.94	94.49	67.82	95.78	94.16	61.89	67.29	51.41	75.53	79.07	55.50	49.36	62.08	87.52	77.89	98.79	86.68	80.48	65.24	89.59	81.31	72.72	76.45
Safe-level-SMOTE	59.35	91.65	91.76	92.80	67.29	92.43	94.88	60.86	67.08	52.24	75.57	80.19	54.48	48.61	61.76										

Table 6

Results for mean ranking of the 10 methods averaged over the 24 datasets and Results for Friedman's test.

Meas.	None Sampling	ROS	SMOTE	BSMOTE	ADASYN	Safe-level-SMOTE	Cluster SMOTE	A-SUWO	MWMOTE	NI-MWMOTE	P-values
Classification method: SVM											
SEN	4.938	5.792	5.354	6.438	4.938	7.083	4.354	8.938	4.917	2.250	2.76E-12**
G-M	5.083	5.542	5.104	7.167	5.042	6.854	4.479	9.063	4.875	1.792	8.02E-15**
F-M	5.375	5.729	5.188	7.375	5.417	6.604	4.625	7.875	4.854	1.958	5.59E-11**
AUC	5.542	5.833	5.250	7.333	5.104	6.583	4.604	8.583	4.604	1.563	5.07E-14**
Classification method: ANN											
SEN	9.167	5.125	4.583	4.438	5.250	5.500	5.354	6.292	6.083	3.208	4.32E-09**
G-M	9.000	4.542	4.604	5.417	6.167	5.208	5.729	5.750	5.875	2.708	2.00E-09**
F-M	7.333	4.417	5.083	6.000	7.042	5.417	5.229	5.833	5.500	3.146	8.13E-05**
AUC	8.042	4.604	4.792	5.625	6.188	5.521	5.938	5.813	5.833	2.646	1.37E-06**
Classification method: RF											
SEN	9.083	6.667	3.750	5.583	3.646	6.125	4.333	7.708	6.458	1.646	3.58E-20**
G-M	8.438	6.771	3.625	6.125	3.458	6.125	4.479	8.042	6.271	1.667	2.62E-19**
F-M	6.729	5.667	4.042	6.604	4.750	5.438	5.188	7.500	6.479	2.604	1.20E-07**
AUC	7.521	6.667	4.021	6.625	4.000	6.146	4.854	7.396	5.958	1.813	6.16E-13**
Classification method: KNN											
SEN	8.063	8.000	4.396	4.958	4.438	5.896	4.104	7.833	5.396	1.917	2.61E-16**
G-M	7.833	7.750	4.833	5.854	4.854	5.708	4.833	7.000	4.708	1.625	3.62E-13**
F-M	6.646	6.563	5.375	5.667	5.875	5.521	5.958	6.042	4.375	2.979	8.73E-05**
AUC	7.542	7.292	4.833	6.042	5.396	5.750	5.167	6.708	4.625	1.646	7.41E-11**

Table 7

Results of Holm's test with NI-MWMOTE as the control method.

No.	$\alpha_{0.05}$	Sensitivity		G-mean		F-measure		AUC		
		Sidak alpha	Algorithms	P-values	Algorithms	P-values	Algorithms	P-values	Algorithms	P-values
Classification method: SVM										
1	0.005683	A-SUWO	4.35E-19**	A-SUWO	2.35E-23**	A-SUWO	2.07E-17**	A-SUWO	1.40E-21**	
2	0.006391	Safe-level-SMOTE	2.19E-11**	BSMOTE	1.52E-14**	BSMOTE	1.10E-12**	BSMOTE	6.28E-16**	
3	0.007301	BSMOTE	5.24E-09**	Safe-level-SMOTE	2.57E-13**	Safe-level-SMOTE	3.98E-11**	Safe-level-SMOTE	8.59E-13**	
4	0.008512	ROS	5.35E-07**	ROS	3.08E-08**	ADASYN	1.34E-06**	ROS	6.53E-10**	
5	0.010206	SMOTE	8.60E-06**	None Sampling	9.77E-07**	ROS	1.77E-06**	None Sampling	7.15E-09**	
6	0.012741	ADASYN	0.00010591	SMOTE	9.77E-07**	SMOTE	6.80E-06**	SMOTE	9.56E-08**	
7	0.016952	None Sampling	0.00013406	ADASYN	1.32E-06**	Cluster SMOTE	8.83E-06**	ADASYN	1.78E-07**	
8	0.025321	MWMOTE	0.00013406	MWMOTE	4.21E-06**	None Sampling	3.14E-05**	MWMOTE	4.55E-06**	
9	0.050000	Cluster SMOTE	0.0026838*	Cluster SMOTE	6.29E-05**	MWMOTE	0.00013217	Cluster SMOTE	7.92E-06**	
Classification method: ANN										
1	0.005683	None Sampling	1.50E-14**	None Sampling	5.72E-16**	None Sampling	1.51E-07**	None Sampling	9.33E-12**	
2	0.006391	A-SUWO	3.07E-05**	ADASYN	2.96E-06**	ADASYN	9.25E-07**	ADASYN	3.54E-06**	
3	0.007301	MWMOTE	7.79E-05**	MWMOTE	1.74E-05**	BSMOTE	0.00026739	Cluster SMOTE	1.52E-05**	
4	0.008512	Safe-level-SMOTE	0.0014742*	A-SUWO	2.83E-05**	A-SUWO	0.00058343	MWMOTE	2.42E-05**	
5	0.010206	Cluster SMOTE	0.0031062*	Cluster SMOTE	4.55E-05**	MWMOTE	0.002488*	A-SUWO	3.04E-05**	
6	0.012741	ADASYN	0.0052811*	BSMOTE	0.00022139	Safe-level-SMOTE	0.0034931*	BSMOTE	9.28E-05**	
7	0.016952	ROS	0.010317**	Safe-level-SMOTE	0.0006342*	Cluster SMOTE	0.0066935*	Safe-level-SMOTE	0.00017628	
8	0.025321	SMOTE	0.067179**	SMOTE	0.009978**	SMOTE	0.012348**	SMOTE	0.0044374*	
9	0.050000	BSMOTE	0.096926**	ROS	0.011731**	ROS	0.097595**	ROS	0.011677**	
Classification method: RF										
1	0.005683	None Sampling	8.11E-29**	None Sampling	8.95E-24**	A-SUWO	9.01E-11**	None Sampling	6.43E-15**	
2	0.006391	A-SUWO	1.82E-21**	A-SUWO	8.45E-22**	BSMOTE	1.29E-07**	A-SUWO	9.58E-15**	
3	0.007301	ROS	7.33E-16**	ROS	2.50E-15**	None Sampling	1.71E-07**	ROS	6.50E-12**	
4	0.008512	MWMOTE	4.83E-15**	MWMOTE	3.36E-13**	MWMOTE	2.97E-07**	BSMOTE	1.35E-11**	
5	0.010206	Safe-level-SMOTE	2.97E-13**	BSMOTE	1.87E-12**	ROS	5.70E-05**	Safe-level-SMOTE	6.66E-10**	
6	0.012741	BSMOTE	8.16E-11**	Safe-level-SMOTE	1.87E-12**	Safe-level-SMOTE	0.00013815	MWMOTE	3.60E-09**	
7	0.016952	Cluster SMOTE	7.25E-06**	Cluster SMOTE	3.84E-06**	Cluster SMOTE	0.00048477	Cluster SMOTE	8.95E-06**	
8	0.025321	SMOTE	0.0004085*	SMOTE	0.0012302*	ADASYN	0.0038292*	SMOTE	0.0013096*	
9	0.050000	ADASYN	0.00052891	ADASYN	0.003055**	SMOTE	0.050447**	ADASYN	0.0013096*	
Classification method: KNN										
1	0.005683	None Sampling	1.30E-18**	None Sampling	2.68E-17**	None Sampling	7.44E-06**	None Sampling	4.43E-15**	
2	0.006391	ROS	2.05E-18**	ROS	6.17E-17**	ROS	9.37E-06**	ROS	4.38E-14**	
3	0.007301	A-SUWO	1.24E-17**	A-SUWO	8.75E-14**	A-SUWO	0.00010277	A-SUWO	7.40E-12**	
4	0.008512	Safe-level-SMOTE	1.68E-09**	BSMOTE	2.34E-09**	Cluster SMOTE	0.00018897	BSMOTE	1.71E-09**	
5	0.010206	MWMOTE	1.04E-07**	Safe-level-SMOTE	6.35E-09**	ADASYN	0.0002806*	Safe-level-SMOTE	1.58E-08**	
6	0.012741	BSMOTE	3.21E-06**	ADASYN	2.83E-06**	BSMOTE	0.00072569	ADASYN	1.76E-07**	
7	0.016952	ADASYN	8.90E-05**	SMOTE	3.73E-06**	Safe-level-SMOTE	0.0012498*	Cluster SMOTE	9.77E-07**	
8	0.025321	SMOTE	0.00014803	Cluster SMOTE	3.73E-06**	SMOTE	0.0025005*	SMOTE	8.37E-06**	
9	0.050000	Cluster SMOTE	0.00079158	MWMOTE	8.46E-09**	MWMOTE	0.066366**	MWMOTE	2.95E-05**	

Nekooimehr & Lai-Yuen, 2016). Table 7 presents the adjusted α and the corresponding p-value at a significance level of $\alpha = 0.05$ for all evaluated metrics and classifiers. Analysis of Table 7 will not be difficult to find that vast majority of the null hypotheses are rejected. Even though some (e.g. Sensitivity in ANN) accepted the null hypothesis of $\alpha = 0.05$, they still rejected the null hypothesis of $\alpha = 0.1$. Therefore, NI-MWMOTE is superior to other oversampling algorithms for all four classifiers and all four measures.

In summary, the comprehensive performance of NI-MWMOTE is superior to other eight oversampling algorithms.

4.3.2. Comparison of imbalanced classification under serious conditions

1) Comparison under high noise conditions

To further explain the superiority of NI-MWMOTE when dealing with noise, we selected six datasets with high noise ratios from Table 1 to analysis, and their noise ratios ($k = 5$) are all 20% and above. Where, to highlight the advantages of the proposed noise processing method, m is 1 when the selected datasets are sampled using NI-MWMOTE. Then, according to Tables 2–5, the best performance times of each evaluation measure were calculated in Table 8.

According to Tables 2–5 and Table 8, it is not difficult to find: 1) NI-MWMOTE has won the most times, 2) Only NI-MWMOTE has achieved the best at least two or more measures in each dataset, and 3) According to Tables 2–5, only NI-MWMOTE and SMOTE have no classification failures, while the other 8 algorithms have classification failures (e.g. Leaf, Ecoli3 and Abalone 9–18 datasets). This is because the knn-based method deletes a large number of minority instances that contain important information, which further increases the imbalance ratios of the datasets, especially when the number of minority instances is very limited and the imbalance ratios are large, such as Abalone 9–18 dataset. Therefore, compared with other algorithms (e.g. MWMOTE), NI-MWMOTE has a great advantage when dealing with noise, especially when the number

of minority instances is particularly limited. The reasons for the poor performance of the remaining methods can be attributed to: 1) The sampling algorithm will be severely affected by noise (e.g. ADASYN), and 2) Removing too many noisy points can result in a serious lack of information in minority class (e.g. MWMOTE and A-SUWO). These all will make learning of imbalanced data more difficult.

2) Comparison of addressing both between-class imbalances and within-class imbalances

To further explain the superiority of NI-MWMOTE when addressing both between-class imbalances and within-class imbalances, we selected four datasets with high imbalanced ratios from Table 1 to analysis, and their imbalanced ratios are all 3.22 and above. The reason we selected the datasets with higher imbalance ratios is that such datasets tend to have a high degree of within-class imbalances, so it is very important to identify small sub-clusters and emphasize them by oversampling (Nekooimehr & Lai-Yuen, 2016). Similarly, to highlight the advantages of the proposed AHC-based method and adaptive sub-cluster sizing method, m was 2 or 3 when the selected datasets were sampled using NI-MWMOTE. Then, according to Tables 2–5, the best performance times of each evaluation measure were summarized in Table 9.

According to Table 9, it is not difficult to find: 1) NI-MWMOTE has won the most times, and 2) Only NI-MWMOTE has achieved the best at least three or more measures in each dataset. Therefore, the experimental results suggest that the NI-MWMOTE algorithm is better than MWMOTE and other processing within-class imbalance algorithms (e.g. Cluster SMOTE and A-SUWO). This is because not only can NI-MWMOTE prevent the small sized sub-clusters away from majority instances from being ignored, but also it adaptively determine the size of the minority sub-cluster. Additionally, it inherits the advantages of the standard MWMOTE (weights minority instances based on the Euclidean distance from hard-to-learn minority instances to majority instances). While A-SUWO

Table 8
Results of the selected datasets with high noise rates.

Dataset (Noise ratios)	Meas.	None Sampling	ROS	SMOTE	BSMOTE	ADASYN	Safe-level-SMOTE	Cluster SMOTE	A-SUWO	MWMOTE	NI-MWMOTE
Leaf (22.50%)	SEN	0	0	0	0	0	0	1	0	0	3
	G-M	0	0	0	0	0	0	0	0	0	4
	F-M	0	0	0	0	0	0	1	0	0	3
	AUC	1	0	1	0	0	0	1	1	0	0
Haberman (24.69%)	SEN	0	0	1	0	0	0	3	0	0	0
	G-M	0	0	0	0	2	0	2	0	0	0
	F-M	0	0	0	0	1	0	2	0	0	1
	AUC	0	0	0	0	0	0	0	0	0	4
Ecoli3 (20.00%)	SEN	0	0	0	0	0	0	0	0	0	4
	G-M	0	0	0	0	0	0	0	0	2	2
	F-M	0	0	0	0	0	0	0	0	2	2
	AUC	0	0	0	0	0	0	0	0	2	2
Abalone 9–18 (47.62%)	SEN	0	0	0	0	0	0	0	0	0	4
	G-M	0	0	0	1	0	0	0	0	0	3
	F-M	0	1	0	0	0	0	0	0	0	3
	AUC	0	1	0	0	0	0	0	0	0	3
Ds1.100(31.72%)	SEN	1	0	0	0	0	0	0	0	0	3
	G-M	1	0	0	0	0	0	0	0	0	3
	F-M	2	1	0	0	0	0	0	0	1	0
	AUC	1	0	0	0	0	0	0	0	0	3
Ds1.10 (27.11%)	SEN	1	0	0	1	0	0	0	0	0	2
	G-M	0	2	0	0	0	0	0	0	0	2
	F-M	0	1	0	1	0	1	0	1	0	0
	AUC	0	1	1	0	0	0	0	0	0	2
Total		7	7	3	3	3	1	10	2	7	53

Table 9

Results of the selected datasets when addressing both between-class imbalances and within-class imbalances.

Dataset (Imbalanced ratios)	Meas.	None Sampling	ROS	SMOTE	BSMOTE	ADASYN	Safe-level-SMOTE	Cluster SMOTE	A-SUWO	MWMOTE	NI-MWMOTE
Wpbc (1/3.22)	SEN	0	0	0	0	0	0	0	0	1	3
	G-M	0	0	0	0	0	0	0	0	1	3
	F-M	0	0	0	0	0	0	0	0	1	3
	AUC	0	0	0	0	0	0	0	0	1	3
Lev (1/9.75)	SEN	0	0	0	2	3	0	0	0	1	0
	G-M	0	0	2	0	0	0	0	1	0	1
	F-M	0	0	1	0	0	0	0	1	1	1
	AUC	0	0	2	0	0	0	0	0	2	1
Yeast (1/3.89)	SEN	0	0	0	1	0	0	0	0	0	3
	G-M	0	0	2	0	0	0	0	0	0	2
	F-M	0	1	1	0	0	0	0	0	2	0
	AUC	0	0	1	0	0	1	0	0	0	2
New-thyroid1 (1/5.14)	SEN	0	0	0	0	2	0	0	0	0	3
	G-M	0	0	0	0	0	0	0	0	0	4
	F-M	1	0	0	0	0	0	0	0	0	3
	AUC	0	0	0	0	0	0	0	0	0	4
Page-blocks0 (1/8.79)	SEN	0	0	0	1	1	0	0	0	0	3
	G-M	0	0	0	0	0	0	0	0	0	4
	F-M	0	0	1	0	0	1	0	0	1	1
	AUC	0	0	0	0	0	0	0	0	0	4
Total		1	1	10	4	6	2	0	2	11	48

Table 10

Results of the selected datasets with limited minority instances.

Dataset (minority/majority)	Meas.	None Sampling	ROS	SMOTE	BSMOTE	ADASYN	Safe-level-SMOTE	Cluster SMOTE	A-SUWO	MWMOTE	NI-MWMOTE
Wpbc (47/151)	SEN	0	0	0	0	0	0	0	0	1	3
	G-M	0	0	0	0	0	0	0	0	1	3
	F-M	0	0	0	0	0	0	0	0	1	3
	AUC	0	0	0	0	0	0	0	0	1	3
Segment (36/396)	SEN	0	0	0	0	0	0	1	0	0	3
	G-M	0	1	0	0	0	0	0	0	0	3
	F-M	1	0	1	0	0	0	0	0	0	2
	AUC	0	1	0	0	0	0	0	0	0	3
Parkinsons (48/147)	SEN	0	0	1	0	1	0	0	0	0	2
	G-M	0	0	0	2	0	0	0	0	0	2
	F-M	1	0	0	1	0	0	0	0	0	2
	AUC	0	0	0	2	0	0	0	0	0	2
New-thyroid1 (35/180)	SEN	0	0	0	0	2	0	0	0	0	3
	G-M	0	0	0	0	0	0	0	0	0	4
	F-M	1	0	0	0	0	0	0	0	0	3
	AUC	0	0	0	0	0	0	0	0	0	4
Breast (36/70)	SEN	0	0	1	0	1	0	0	0	0	2
	G-M	0	0	1	0	0	0	0	1	0	2
	F-M	0	0	1	0	0	0	0	1	0	2
	AUC	0	0	0	0	0	0	0	0	0	4
Ecoli3 (35/301)	SEN	0	0	0	0	0	0	0	0	0	4
	G-M	0	0	0	0	0	0	0	0	2	2
	F-M	0	0	0	0	0	0	0	0	2	2
	AUC	0	0	0	0	0	0	0	0	2	2
Abalone 9–18 (42/689)	SEN	0	0	0	0	0	0	0	0	0	4
	G-M	0	0	0	1	0	0	0	0	0	3
	F-M	0	1	0	0	0	0	0	0	0	3
	AUC	0	1	0	0	0	0	0	0	0	3
Ds1.100 (50/1621)	SEN	1	0	0	0	0	0	0	0	0	3
	G-M	1	0	0	0	0	0	0	0	0	3
	F-M	2	1	0	0	0	0	0	0	1	0
	AUC	1	0	0	0	0	0	0	0	0	3
Ds1.10 (50/1621)	SEN	1	0	0	1	0	0	0	0	0	2
	G-M	0	2	0	0	0	0	0	0	0	2
	F-M	0	1	0	1	0	1	0	1	0	0
	AUC	0	1	1	0	0	0	0	0	0	2
Total		9	9	6	8	4	1	1	3	11	93

strives to overcome both between-class imbalances and within-class imbalances, knn method removes too many minority instances, which causes minority classes to miss more instances with important information. Additionally, overly complex clustering process of A-SUWO is more likely to cause over-fitting.

3) Comparison under limited minority instances

In general, the fewer the minority instances, the less information the classifiers can obtain, which makes imbalanced learning more difficult. Especially when datasets are noisy and there are within-class imbalances, imbalanced learning becomes extremely difficult. This is because the limited data will be sparsely distributed, which is likely to exacerbate the generation of noise and within-class imbalances. Therefore, to further explain the superiority of NI-MWMOTE when encountering datasets containing limited minority instances, we selected nine datasets with limited minority instances from Table 1 to analysis, and the total number of their minority instances does not exceed 50. Then, according to Tables 2–5, the best performance times of each evaluation measure were calculated in Table 10.

According to Tables 2–5 and Table 10, it is not difficult to find: 1) NI-MWMOTE has won the most times, 2) Only NI-MWMOTE has achieved the best at least three or more measures in each dataset, and 3) According to Tables 2–5, only NI-MWMOTE has no classification failures, while the other 9 algorithms all have at least one failure (e.g. Parkinsons, Heart, Ecoli3 and Abalone 9–18 datasets). Therefore, it is not difficult to conclude that NI-MWMOTE can fully utilize limited minority instances to mine a larger amount of information for classification. This is firstly due to our reasonable noise processing strategy, and secondly due to the AHC-based method and adaptive sub-cluster sizing method and MWMOTE's own method to deal with within-class imbalances. Additionally, NI-MWMOTE inherits the advantages of standard MWMOTE to synthesize a wide range of minority instances, which will make the sparse instances as a whole tighter. MWMOTE's second best performance justifies this.

4.3.3. NI-MWMOTE's computational cost and its simplification

According to the analysis in Sections 4.3.1 and 4.3.2, it can be known that NI-MWMOTE has performed the best overall. And it has a great advantage in dealing with noise, solving both between-class imbalances and within-class imbalances, and using limited minority data for learning. However, compared with standard MWMOTE, NI-MWMOTE increases the complexity of the algorithm, and most of its execution time is spent on selecting the optimal noise processing scheme. This is because: 1) NI-MWMOTE increases the number of noise to be processed one by one (i.e., the step size is 1.), which will greatly increase the solution cost of datasets with high noise ratios, and 2) The judgment of the reasonableness of noise processing is performed multiple times, which also increases the computational cost.

Therefore, in this section, we not only show the solution cost of NI-MWMOTE, but also propose a scheme to simplify the computational cost of it. The method we intend to adopt is as follows: 1) Increase the step size of noise processing, that is, set the step size to $S = \text{round}(\text{length}(\text{SNp}) / i)$, $i = 1, 2, \dots, \text{length}(\text{SNp}(\text{suspected noise of minority instances}))$, and 2) Reduce the number of times to determine the reasonableness of each strategy ($q = 1$), that is, the SMOTE algorithm here is executed only once.

To verify the above ideas, we selected three datasets with high noise ratios from Table 1 (i.e., Abalone 9–18, Ds1.100 and Ds1.10 datasets) for exploration. The reason we chose these three datasets is that Abalone 9–18 can represent small-scale imbalanced datasets, Ds1.10 can represent larger-scale low-dimensional imbalanced datasets, and Ds1.100 can represent larger-scale High-latitude imbalanced datasets. We selected SVM and KNN for classification experiments, and used G-mean as the evaluation measure. And all the parameters' settings are the same as above. Additionally, the method of obtaining results is the same as above. Finally, the results are summarized in Table 11.

According to Table 11, it is not difficult to find: 1) The proposed standard NI-MWMOTE algorithm is much more expensive than MWMOTE and A-SUWO algorithms, especially when it comes to larger-scale High-latitude imbalanced datasets. 2) With the

Table 11
Solving costs and results of NI-MWMOTE under different conditions.

Datasets	Methods	G-M (%)	Time (s)	Datasets	Methods	G-M (%)	Time (s)	Datasets	Methods	G-M (%)	Time (s)
Abalone 9–18 (SVM)	A-SUWO	2.50	30.33	Ds1.10 (SVM)	A-SUWO	68.86	124.26	Ds1.100 (SVM)	A-SUWO	62.53	268.90
	MWMOTE	27.34	8.50		MWMOTE	69.28	24.33		MWMOTE	72.86	8.21
	$i = 1, q = 1$	59.64	47.44		$i = 1, q = 1$	73.99	163.68		$i = 1, q = 1$	73.63	125.32
	$i = 2, q = 1$	65.17	73.07		$i = 2, q = 1$	73.89	187.16		$i = 2, q = 1$	72.98	134.01
	$i = 3, q = 1$	60.84	67.63		$i = 3, q = 1$	71.78	213.97		$i = 3, q = 1$	73.86	166.63
	$i = 4, q = 1$	59.21	144.85		$i = 4, q = 1$	71.79	245.38		$i = 4, q = 1$	73.00	191.75
	$i = 5, q = 1$	60.51	100.55		$i = 5, q = 1$	73.00	287.64		$i = 5, q = 1$	73.91	158.15
	$i = 6, q = 1$	61.05	180.66		$i = 6, q = 1$	73.15	312.67		$i = 6, q = 1$	72.84	186.16
	$i = 7, q = 1$	63.06	210.48		$i = 7, q = 1$	73.62	375.42		$i = 7, q = 1$	73.87	210.25
	$i = 8, q = 1$	60.19	209.75		$i = 8, q = 1$	72.20	388.66		$i = 8, q = 1$	72.86	226.15
	$i = 9, q = 1$	56.13	223.67		$i = 9, q = 1$	74.01	462.35		$i = 9, q = 1$	73.06	250.05
	$i = 10, q = 1$	57.41	150.94		$i = 10, q = 1$	71.01	504.22		$i = 10, q = 1$	73.56	254.64
	$S = 1, q = 1$	63.43	308.82		$S = 1, q = 1$	73.23	750.96		$S = 1, q = 1$	73.96	709.11
	$S = 1, q = 5$	62.69	1154.00		$S = 1, q = 5$	73.86	3283.25		$S = 1, q = 5$	73.94	3480.56
Abalone 9–18 (KNN)	A-SUWO	23.47	27.43	Ds1.10 (KNN)	A-SUWO	66.47	87.68	Ds1.100 (KNN)	A-SUWO	77.14	336.22
	MWMOTE	46.68	1.36		MWMOTE	71.78	3.16		MWMOTE	80.64	4.97
	$i = 1, q = 1$	63.68	20.46		$i = 1, q = 1$	77.25	59.92		$i = 1, q = 1$	79.54	78.86
	$i = 2, q = 1$	64.75	20.75		$i = 2, q = 1$	76.78	62.85		$i = 2, q = 1$	79.48	99.05
	$i = 3, q = 1$	66.04	25.37		$i = 3, q = 1$	77.44	64.31		$i = 3, q = 1$	79.82	115.50
	$i = 4, q = 1$	65.96	32.41		$i = 4, q = 1$	77.07	81.89		$i = 4, q = 1$	80.22	112.89
	$i = 5, q = 1$	70.09	46.57		$i = 5, q = 1$	78.10	95.42		$i = 5, q = 1$	80.04	121.75
	$i = 6, q = 1$	66.43	48.39		$i = 6, q = 1$	77.12	95.99		$i = 6, q = 1$	79.80	132.85
	$i = 7, q = 1$	65.36	87.81		$i = 7, q = 1$	76.03	200.67		$i = 7, q = 1$	80.09	135.63
	$i = 8, q = 1$	66.37	70.74		$i = 8, q = 1$	76.60	168.71		$i = 8, q = 1$	79.71	139.93
	$i = 9, q = 1$	69.08	67.89		$i = 9, q = 1$	76.08	172.11		$i = 9, q = 1$	79.12	155.99
	$i = 10, q = 1$	66.70	66.97		$i = 10, q = 1$	77.98	176.19		$i = 10, q = 1$	80.00	184.61
	$S = 1, q = 1$	66.35	111.30		$S = 1, q = 1$	77.90	259.37		$S = 1, q = 1$	79.82	448.59
	$S = 1, q = 5$	72.30	509.74		$S = 1, q = 5$	77.17	817.00		$S = 1, q = 5$	81.19	2044.24

increase of the step size and the reducing of the number of SMOTE's execution times, the solution cost is greatly reduced, and the accuracies of the solution have downward trends, but not obviously. Therefore, this does not significantly weaken the sampling advantage of NI-MWMOTE, that is, it is still significantly better than the other algorithms. 3) For all KNN's datasets and some SVM's datasets (e.g. Ds1.100), the costs of simplified NI-MWMOTE are much smaller than those of A-SUWO, especially when dealing with larger-scale High-latitude datasets. For the rest of the SVM's datasets, the solution cost of NI-MWMOTE is higher than the solution cost of A-SUWO, but the difference is not obvious. Therefore, when the user pursues higher precision and ignores the solution cost, a processing method with a small step size and SMOTE execution multiple times can be selected. On the contrary, when the user pursues a low solution cost, a strategy with a large step size and fewer executions of SMOTE can be selected.

5. Conclusions

Recently, many oversampling methods have been presented for imbalanced binary classification problems. While these algorithms have achieved certain results in dealing with imbalanced datasets, the following problems still exist:

- 1) Most existing algorithms cannot accurately and effectively deal with noise, especially the processing methods based on pure knn.
- 2) Most algorithms can not address both between-class imbalances and within-class imbalances effectively, especially the small minority sub-clusters that are easy to ignore.
- 3) Some algorithms achieve their respective improvements at the cost of high complexity, leading to them having certain limitations in practical applications, and when the parameters are not well adjusted, it is easy to cause over-fitting.

Therefore, we proposed an improving noise-immunity majority weighted minority oversampling technique (NI-MWMOTE) to address the above three problems. Compared to other algorithms (e.g. MWMOTE and A-SUWO, etc.), the advantages of NI-MWMOTE can be summarized:

- a. Adaptively remove noise. NI-MWMOTE calculates the probability of suspected noise as true noise by considering the Euclidean distance from suspected noise to class center and its neighbor density. Then, true noise will be adaptively removed by iteration according to the above probabilities and the misclassification error.
- b. Use unsupervised AHC to prevent ignoring small minority sub-clusters. The combination of AHC and standard MWMOTE can more effectively resolve both between-class imbalances and within-class imbalances. Moreover, AHC is simpler and easier to implement than the semi-supervised cluster of A-SUWO, and the simplified NI-MWMOTE method just proved this aspect. Additionally, fewer sub-clusters are less likely to cause over-fitting.
- c. Adaptive Sub-cluster Sizing. NI-MWMOTE determines the sub-cluster sizes adaptively using the standardized average error rate and cross-validation such as A-SUWO.

To test the effectiveness and superiority of NI-MWMOTE, we performed classification experiments using four popular classifiers and 24 publicly available datasets with different dimensions and imbalance ratios. And we compared it with other eight popular oversampling algorithms. Results suggest that NI-MWMOTE performs significantly better in most imbalanced datasets compared

to other algorithms, especially in the datasets with higher noise ratios, higher imbalanced ratios and limited minority instances. Additionally, as an improved version of MWMOTE, while standard NI-MWMOTE or simplified NI-MWMOTE adds complexity to the algorithm, it is extremely necessary. This is because compared to MWMOTE, the sampling performance of it has been greatly improved in most imbalanced datasets, and its improvement effect is less affected by parameters.

As future works, the complexity of NI-MWMOTE will be further simplified, making it more suitable for solving large-scale imbalanced classification problems. Additionally, in this paper, we have not used NI-MWMOTE to explore the classification tasks not modelled by feature vectors. Perhaps, in the future, we will shift our focus to this type of task based on the idea of NI-MWMOTE. That is, we will replace the SMOTE idea in the existing algorithms (e.g. INOS (Cao et al., 2013) and MBOS (Gong & Chen, 2016), etc.) with the NI-MWMOTE idea or explore a new NI-MWMOTE method based on the Edit Distance (Castellanos et al., 2018) to address the problems of noise, between-class imbalances and within-class imbalances at the same time. After all, our proposed method has achieved great success in the classification tasks modelled by feature vectors.

CRedit authorship contribution statement

Jianan Wei: Conceptualization, Methodology, Software, Investigation, Writing - original draft. **Haisong Huang:** Resources, Validation, Formal analysis, Visualization, Software, Data curation, Supervision. **Liguo Yao:** Validation, Formal analysis, Visualization, Supervision. **Yao Hu:** Validation, Formal analysis, Visualization. **Qingsong Fan:** Writing - review & editing, Supervision. **Dong Huang:** Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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